

**DISTRICT-LEVEL PATTERNS OF LAND
FRAGMENTATION, IRRIGATION ACCESS, AND
AGRICULTURAL PRODUCTION SUFFICIENCY IN
NEPAL**

A Thesis

Submitted to the Department of Economics, Patan Multiple Campus,
Faculty of Humanities and Social Sciences, Tribhuvan University, Nepal, in
Partial Fulfillment of the Requirements of the Degree of

**MASTER OF ARTS
in
ECONOMICS**

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July 2026

DECLARATION

I hereby declare that this thesis entitled “DISTRICT-LEVEL PATTERNS OF LAND FRAGMENTATION, IRRIGATION ACCESS, AND AGRICULTURAL PRODUCTION SUFFICIENCY IN NEPAL” which I have submitted to the Department of Economics, Patan Multiple Campus, in partial fulfillment of the requirements for the Degree of MASTER OF ARTS in ECONOMICS, is entirely my original work prepared under the guidance of my supervisor. I have made due acknowledgements to all ideas and information borrowed from different sources in the course of writing this thesis. The results of this thesis have not been presented or submitted anywhere else for the award of any degree. I shall be solely responsible for any evidence found against my declaration.

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This thesis entitled DISTRICT-LEVEL PATTERNS OF LAND FRAGMENTATION, IRRIGATION ACCESS, AND AGRICULTURAL PRODUCTION SUFFICIENCY IN NEPAL has been prepared by Mr. Sujil Devkota under my guidance and supervision. I, hereby, recommend it in partial fulfillment of the requirements for the Degree of MASTER OF ARTS in ECONOMICS for final examination.

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LETTER OF APPROVAL

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my thesis supervisor, Dr. Kumar Bhattarai, Assistant Professor, Department of Economics, Patan Multiple Campus, for his continuous guidance, constructive feedback, and encouragement throughout this study. I am thankful to the Head of the Department and the faculty members of the Department of Economics for their valuable suggestions and academic support.

I gratefully acknowledge the National Statistics Office, Government of Nepal, for making the district tables of the National Sample Census of Agriculture 2021/22 publicly available, which form the empirical foundation of this thesis. Finally, I am deeply indebted to my family and friends for their constant support and motivation during the course of this work. Any errors or shortcomings that remain are entirely my own.

Sujil Devkota

July 2026

ABSTRACT

This thesis examines how land fragmentation and irrigation access are associated with agricultural production sufficiency across the 77 districts of Nepal. Its objectives are to document the district-level patterns of land fragmentation and irrigation access, to document the district-level patterns of agricultural production sufficiency, and to analyse the relationship between them, including the moderating role of irrigation. A district-level dataset is constructed from the official tables of the National Sample Census of Agriculture (NSCA) 2021/22. Because the outcome is a bounded proportion, the analysis combines descriptive and correlation evidence with a weighted ordinary least squares benchmark and a fractional logit model with province fixed effects, mean-centred interactions, and extensive robustness checks.

The data reveal large spatial variation: average parcels per holding range from 1.5 to 6.3, the irrigated-area share from about 3 to 91 percent, and the insufficiency share from 32 to 87 percent. In the multivariate results, irrigation coverage is the dominant correlate: its association with insufficiency is negative and statistically significant in every specification, attenuating but remaining significant when ecological-belt controls replace province effects. Predicted insufficiency differs by roughly 20 percentage points between low- and high-irrigation profiles (a descriptive contrast, not an intervention effect), with bootstrap intervals that exclude zero. The association with land fragmentation is absorbed by geography once fixed effects and controls are included, and the fragmentation–irrigation interaction is statistically null. The severe, 10–12-month food-insufficiency measure is treated as a distinct dimension of deprivation.

The findings are district-level associations rather than causal effects, and irrigation placement is not random, so the evidence is read as conditional on Nepal’s agro-ecological geography. The pattern is nonetheless consistent with prioritising irrigation expansion in low-coverage hill and mountain districts, pursuing land consolidation only as a complement bundled with irrigation and mechanisation, and targeting chronic food deprivation toward near-landless peri-urban and Terai holdings.

Key words: land fragmentation, irrigation access, production sufficiency, Nepal

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ACRONYMS AND ABBREVIATIONS

APE	Average Partial Effect
FE	Fixed Effects
FMIS	Farmer-Managed Irrigation System
GDP	Gross Domestic Product
ha	Hectare
MoF	Ministry of Finance
NSCA	National Sample Census of Agriculture
NSO	National Statistics Office
OLS	Ordinary Least Squares
SE	Standard Error
TU	Tribhuvan University
VIF	Variance Inflation Factor

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

Agriculture remains the backbone of the Nepalese economy and the primary basis of livelihood for the majority of rural households. Although the economy has gradually shifted toward services, the sector still contributes about one-fourth of the country's gross domestic product (GDP) (Ministry of Finance [MoF], 2024), and more than two-thirds of all agricultural holdings (about 70 percent) report agriculture as their main source of income (NSO, 2023). For most rural households, farming is not only an economic activity but also the foundation of food security, employment, and social organisation, so that the structural condition of agriculture has direct implications for poverty, nutrition, and overall household welfare.

This dependence persists even as the economy transforms. The share of agriculture in GDP has declined gradually over the past two decades while the sector continues to employ the majority of the workforce, an imbalance that signals stagnant agricultural productivity relative to the rest of the economy; at the same time, the country's food import bill has grown, and successive Economic Surveys note that domestic production has not kept pace with domestic demand (MoF, 2024). The condition of agriculture is therefore increasingly framed in policy debate not only as a rural welfare issue but as a macroeconomic one, and the capacity of farm households to feed themselves from their own land is the most basic indicator of the sector's health.

However, Nepalese agriculture operates under several long-standing structural constraints. Holdings are small, parcels are scattered, irrigation is unevenly distributed, market connectivity is weak, and farming is increasingly exposed to climate-related risk. The National Sample Census of Agriculture (NSCA) 2021/22 records more than four million agricultural holdings, with an average holding size of only about 0.54 hectare (ha), typically divided into roughly 2.8 separate parcels (National Statistics Office [NSO], 2023). Such small and fragmented holdings raise the time spent travelling between plots, increase boundary losses, complicate supervision, and make mechanisation

and coordinated water management more difficult.

The same census confirms that agriculture remains central to household survival even as it struggles to meet household food needs. While agriculture is the main income source for most holdings, only about 45 percent report that their own agricultural production is sufficient for household consumption, leaving roughly 55 percent who report insufficiency (NSO, 2023). This gap shows that the structural condition of farming is not merely a production concern but also a welfare concern: the reported inability of a holding to meet household consumption from its own production is a practical indicator of agricultural strain, even though it measures production relative to household needs rather than poverty or food security as such.

Irrigation is one of the most decisive factors shaping this outcome. Nationally, about 54.5 percent of the total area of agricultural holdings is irrigated, but coverage is highly uneven across districts, ranging from as little as 2.9 percent in Rukum East to as high as 91.1 percent in Bara (NSO, 2023). Because irrigation reduces dependence on the monsoon, supports multiple cropping, and enables the use of improved inputs, districts with wider irrigation access are expected to convert their land and labour into adequate production more effectively than rainfed districts. Land fragmentation displays a similarly wide spread: average parcels per holding range from about 1.5 in Lalitpur to 6.3 in Humla, while the share of holdings reporting insufficient production ranges from about 32 percent in Rupandehi to about 87 percent in Dolpa.

The geography of these constraints is not accidental. Nepal's surface irrigation is concentrated in the Terai plains, where the terrain, the rivers descending from the mountains, and a long history of public canal investment make gravity-fed systems feasible, while hill and mountain agriculture depends on smaller farmer-managed systems and on rainfall. Fragmentation, for its part, is reproduced by partible inheritance, under which holdings are subdivided among heirs in each generation (B. N. Dhakal & Khanal, 2018), and it is most severe where cultivable land is scarcest, in the high hills and mountains. Structural disadvantage therefore accumulates: the districts with the most fragmented holdings are typically also the districts with the least irrigation, a correlation that any analysis of either condition must confront directly.

This combination of severe structural constraints and wide district-level variation makes

Nepal a particularly suitable setting for studying how land fragmentation and irrigation access relate to agricultural production sufficiency. Although fragmentation and irrigation are frequently invoked in Nepalese agricultural policy, there is limited nationwide empirical work that examines them together using the most recent agriculture census. This study therefore constructs a district-level dataset from the official NSCA 2021/22 tables and analyses how land fragmentation and irrigation access are associated with agricultural production sufficiency across all 77 districts of Nepal.

1.2 Statement of the Problem

Land fragmentation and irrigation access are widely recognised in the international literature as important determinants of agricultural productivity, efficiency, and food security. A large body of evidence indicates that fragmented holdings raise production and transaction costs and lower technical efficiency, because scattered plots impede mechanisation and increase travel, supervision, and water-management costs (Ali et al., 2019; Hao et al., 2023; Manjunatha et al., 2013; Niroula & Thapa, 2005). At the same time, access to irrigation has repeatedly been shown to raise yields and incomes and to reduce household food insecurity in developing countries (Ebrahim & Toy, 2024; Jambo et al., 2021; Kafle & Balasubramanya, 2022). Taken together, this literature suggests that the structure of landholdings and the availability of irrigation jointly shape whether smallholder farming can generate enough output for household consumption. Because scattered parcels raise the costs of supervising plots and of managing water across them, the disadvantage associated with fragmentation may also be weaker, or stronger, where irrigation is assured, which makes the interaction between the two conditions an empirical question in its own right.

In Nepal, these two conditions are especially pronounced. Holdings are among the smallest and most fragmented in South Asia (NSO, 2023; Niroula & Thapa, 2005), and irrigation coverage varies enormously across districts. Despite agriculture being the dominant source of income, roughly 55 percent of holdings report that their agricultural production is insufficient for household consumption, and the burden of insufficiency is concentrated in the hill and mountain districts, with the highest province-level averages in Karnali and Gandaki (NSO, 2023). This points to a serious and spatially uneven

problem of agricultural production insufficiency that is plausibly linked to the structural conditions of fragmentation and limited irrigation.

Existing Nepalese research has examined many related themes, including the causes and consequences of fragmentation (B. N. Dhakal & Khanal, 2018), land underutilisation and abandonment (U. Khanal, 2018; Ojha et al., 2017; Paudel et al., 2025), irrigation access and governance (Nepal et al., 2024; Pariyar et al., 2018; Shrestha et al., 2023), the determinants of household food security (Joshi & Joshi, 2017; Karki et al., 2015), and the effects of migration and feminisation on farm households (Kim et al., 2019; Pandey et al., 2021). However, most of these studies are local case studies, single-scheme analyses, or assessments of one factor at the household level. There is still very limited nationwide, district-level evidence that examines land fragmentation and irrigation access jointly and links them to agricultural production sufficiency using the most recent agriculture census. This is the gap that the present study addresses.

The scale of the variation involved underlines why a nationwide, district-level analysis is needed. Irrigation coverage differs across districts by a factor of more than thirty, and the insufficiency share by a factor of almost three; problems of this spatial magnitude cannot be characterised, let alone addressed, from single-district case studies. At the same time, the three conditions are visibly entangled, with the most fragmented, least irrigated, and most insufficient districts overlapping heavily, so their separate associations with sufficiency can only be distinguished in a multivariate framework that holds location constant.

Accordingly, the study seeks to answer the following research questions:

- i. How do land fragmentation and irrigation access vary across the districts of Nepal?
- ii. How does agricultural production sufficiency vary across the districts of Nepal?
- iii. What is the district-level relationship between land fragmentation, irrigation access, and agricultural production sufficiency, and does irrigation moderate the disadvantage associated with fragmentation?

1.3 Objectives of the Study

The general objective of the study is to examine the district-level patterns and relationships of land fragmentation, irrigation access, and agricultural production sufficiency in Nepal. In line with the three research questions, the specific objectives are:

- i. To examine the district-level patterns of land fragmentation and irrigation access across the districts of Nepal;
- ii. To examine the district-level patterns of agricultural production sufficiency across the districts of Nepal; and
- iii. To analyse the district-level relationship between land fragmentation, irrigation access, and agricultural production sufficiency in Nepal, including whether irrigation access moderates the association between fragmentation and sufficiency.

1.4 Hypotheses of the Study

Guided by the farm-household framework developed in Chapter III and the empirical evidence reviewed in Chapter II, and stated in terms of the insufficiency outcome that the study models, the thesis tests the following hypotheses about the district-level associations:

- H1. Districts with wider irrigation coverage report lower agricultural production insufficiency.
- H2. Districts with more fragmented holdings report higher agricultural production insufficiency.
- H3. Irrigation moderates the association between fragmentation and insufficiency, with the direction left open: irrigation and land consolidation may act as substitutes, so that assured irrigation softens the fragmentation disadvantage, or as complements, so that the disadvantage is larger where irrigation is limited.

All three are hypotheses about conditional district-level associations, given the controls and the fixed effects of the empirical model, and not about household-level causal effects;

the identification threats discussed in Chapter III apply to each. The formal statement of each hypothesis in terms of the coefficients of the estimating equation is given in Section 3.7.

1.5 Significance of the Study

This study is significant for both academic and policy reasons. From an academic standpoint, it contributes to scientific knowledge by addressing a clear gap in the Nepalese literature: it brings together three connected issues, namely land fragmentation, irrigation access, and agricultural production sufficiency, within a single empirical framework, using the most recent nationwide agriculture census. By constructing a district-level dataset for all 77 districts from official tables and applying transparent and reproducible statistical and econometric methods, the study offers evidence with broader national relevance than the local case studies that dominate the existing literature. It also establishes a replicable research workflow that future studies can extend with additional controls or with household-level microdata when these become available.

From a policy standpoint, the study can help identify where the structural disadvantages of Nepalese agriculture are most severe and which conditions are most strongly associated with agricultural production insufficiency. By documenting the district-level variation in fragmentation, irrigation, and sufficiency, and by estimating their associations, the findings can inform the targeting of irrigation expansion, land management and consolidation policy, subsidy and credit programmes, and the prioritisation of districts for agricultural support. Such evidence is directly relevant to national objectives of improving food security and rural welfare.

1.6 Scope and Limitations of the Study

The study is national in coverage but district in resolution. It uses the district-level tables of the NSCA 2021/22 for all 77 districts of Nepal and focuses on the relationship between land fragmentation, irrigation access, and agricultural production sufficiency, while controlling for average holding size, agricultural loans, government subsidy, mechanisation, market access, and climate impact. Throughout the thesis, production sufficiency is

analysed through its complement, the share of holdings reporting that their own production is insufficient for household consumption, and estimated signs are interpreted accordingly. The analysis is confined to the variables consistently reported in the official census and does not attempt to cover every dimension of rural welfare.

The study is not free from limitations, the main ones being:

- i. The analysis uses cross-sectional data from a single census round, so it documents associations rather than household-level causal effects.
- ii. The unit of analysis is the district; the study therefore captures broad spatial patterns rather than individual farm behaviour and is subject to the usual limits of ecological inference.
- iii. The sufficiency measure is the holder's self-reported assessment in the census; it is a perception-based indicator of production relative to household needs and may differ from objectively measured production–consumption gaps. Because it is judged against household needs, the district average also mixes a production effect with an average-household-size composition effect; a direct control for average household size (Section 4.4.3) indicates that this composition effect does not drive the study's results.
- iv. Fragmentation is proxied by the average number of parcels per holding; parcel dispersion and the distance between plots are not observed in the census tables.
- v. Some potentially relevant district characteristics, such as rainfall, soil quality, crop mix, remittance dependence, and road quality, are not available in the same official dataset, although broad agro-ecological differences are captured by the belt and province controls.
- vi. The district shares are themselves estimates from the census sample rather than error-free observations; this sampling uncertainty in the regressors and the outcome is acknowledged but not propagated through the models.
- vii. The census reference year is 2021/22, a period that partly coincided with disruptions related to COVID-19, and structural conditions may have evolved since.

- viii. The study concentrates on agricultural production sufficiency and closely related agricultural conditions rather than on the full range of food-security and welfare outcomes.

1.7 Outline of the Study

The thesis is organised into five chapters. Chapter I introduces the study, presenting the background, statement of the problem, objectives, hypotheses, significance, scope and limitations, and the overall structure. Chapter II reviews the international and national literature on land fragmentation, irrigation, and agricultural production sufficiency, reviews the relevant policy context, and identifies the research gap. Chapter III explains the research methodology, including the research design, the theoretical and conceptual framework, the nature and sources of data, the study area, the econometric model, the definitions of the key terms, and the operational definitions of the variables. Chapter IV presents the data and analyses the results, moving from the district-level patterns to the multivariate estimates, robustness checks, diagnostic tests, and discussion. Chapter V summarises the study, draws conclusions, and offers the recommendations of the study.

CHAPTER II

REVIEW OF LITERATURE

2.1 Introduction

This chapter reviews the theoretical and empirical literature relevant to the relationship between land fragmentation, irrigation access, and agricultural production sufficiency. Conceptually, land fragmentation refers to a situation in which a single holding is operated as several spatially separated parcels rather than as one consolidated block. Although fragmentation can offer minor benefits, such as spreading agro-climatic risk across plots, the dominant concern in the literature is that it raises travel, supervision, and coordination costs and obstructs mechanisation and water management (Niroula & Thapa, 2005). Irrigation, by contrast, is one of the clearest productivity-enhancing inputs available to smallholders, because it reduces dependence on rainfall, enables multiple cropping, and supports the use of improved seeds and fertiliser. Agricultural production sufficiency, the outcome of interest in this study, refers to whether a holding produces enough for its household's consumption and is therefore a practical and policy-relevant welfare indicator. Throughout the thesis, the term *agricultural production sufficiency* denotes this construct, while *food insufficiency* is reserved for the census measure of the duration of household food shortage; the full theoretical and conceptual framework of the study is developed in Section 3.3. The chapter first sets out the theoretical review, then reviews the international literature and the national (Nepalese) literature, in both cases moving from the most recent studies to the earliest, then reviews the domestic policy context, and finally synthesises these to identify the research gap that this study addresses.

2.2 Literature Review

2.2.1 Theoretical Review

The natural theoretical home for a production-sufficiency outcome is the agricultural household model, formalised in the volume edited by Singh et al. (1986) and surveyed by Taylor and Adelman (2003). In this framework the farm household is simultaneously a producer, a consumer, and a supplier of its own labour, and it maximises utility subject to production, time, and income constraints. When all markets work perfectly, production decisions separate from consumption decisions: the household first maximises farm profit and then buys or sells food as needed, so whether its own production covers its consumption is economically irrelevant. When markets for food, labour, credit, or insurance are thin or missing, as they are in much of rural Nepal, the two decisions become non-separable, and the household's own production becomes a direct determinant of its food consumption. The census question underlying this thesis, whether a holding's own production was sufficient for household consumption, is a meaningful welfare indicator precisely under such non-separability.

Within this framework, land fragmentation operates through transaction and supervision costs. Scattered parcels raise the time cost of travel between plots, the cost of supervising family and hired labour, and the losses along boundaries, and they obstruct lumpy investments such as machinery and tubewells whose returns depend on contiguous area (Niroula & Thapa, 2005). Against these costs stands a risk-diversification benefit: plots spread across different micro-environments insure the household against localised weather, pest, and soil shocks. Which force dominates is an empirical question, and the answer plausibly depends on how binding the household's labour constraint is. This is why recent evidence that the fragmentation penalty operates mainly through labour productivity (Hao et al., 2023) is so relevant to out-migration economies such as Nepal.

Irrigation enters the household model as a risk-reducing and productivity-enhancing input. Reliable water weakens the dependence of output on the monsoon, permits multiple cropping cycles on the same land, and raises the marginal return to complementary inputs such as improved seed and fertiliser (Mupaso et al., 2023). For a quasi-subsistence

household, irrigation therefore raises both the level and the reliability of own production relative to consumption needs, and at the district level wider irrigation coverage should translate into a smaller share of holdings reporting insufficiency.

These theoretical elements, namely non-separability, the transaction-cost and risk-diversification margins of fragmentation, and irrigation as a risk-reducing input, are operationalised in the conceptual framework of Section 3.3, and they motivate the variables, the interaction term, and the geographic controls of the empirical model.

2.2.2 Empirical Review

The empirical literature is reviewed in two strands: international studies that establish the broad relationships between land fragmentation, irrigation, and agricultural performance, and national studies that place these relationships in Nepal's own agrarian context.

2.2.2.1 International Context

The international literature gives this study its broader analytical foundation. It covers four connected themes: the consequences of land fragmentation for farm performance and food security, the gains from land consolidation, the role of irrigation in productivity and food security, and the wider determinants of household food security. The studies are presented from the most recent year to the earliest.

Nyamba and Zidouemba (2025) calculate a net income gain of roughly 1,200 US dollars per hectare for irrigating households in Burkina Faso. Kiratu et al. (2025) compare farmer-led and public irrigation schemes in Kenya and find that both significantly raise agricultural income and household dietary diversity, though partly at the cost of reduced on-farm production diversity. Using panel data from Bangladesh, Shifat et al. (2025) report that asset holdings improve household food security.

Ebrahim and Toy (2024) use propensity-score matching for Ethiopia and find that participation in small-scale irrigation significantly raises household income, food expenditure, and food security, with gains of around 55 percent in both income and food expenditure for irrigators.

Hao et al. (2023) apply an instrumental-variable approach to microdata on rice, wheat, and maize farmers across 14 provinces of China. They estimate that a 1 percent increase in fragmentation lowers labour productivity by roughly 20 to 22 percent, mainly by impeding the use of machinery. The same study finds small positive effects of fragmentation on rice land productivity, so the dominant penalty operates through labour rather than land, a channel that is especially relevant in labour-scarce, out-migration settings such as Nepal. Janus et al. (2023) show across selected European and Asian countries that smaller, irregular fields and longer farm-to-plot distances raise cultivation costs and fuel consumption, lowering the competitiveness of agriculture. Zheng et al. (2023) find in rural China that fragmentation encourages cropland abandonment, a further channel through which it can undermine food supply. On consolidation, Do et al. (2023) report with household panel data that consolidation enhances rice productivity in Vietnam and reallocates labour from farm to non-farm work. Nsabimana et al. (2023) show for Rwanda that the national Land Use Consolidation programme increased the uptake of improved seeds, inorganic fertiliser, pesticides, and irrigation among smallholders in both the short and the long run. On irrigation, a systematic review by Mupaso et al. (2023) finds the evidence on smallholder irrigation to be context-dependent but concludes that investment in irrigation remains a key strategy for raising productivity and reducing rural poverty. Wudil et al. (2023) find for an irrigation scheme in Nigeria that irrigation access, asset possession, land size and quality, and education improve household food security.

Phan et al. (2022) estimate with panel data for rice-farming households in Vietnam that a 1 percent increase in a land fragmentation index raises the probability that a household experiences the most severe level of food insecurity by about 4.8 percent, while larger cultivated area, credit access, and savings reduce it. D. Tran et al. (2022) evaluate a plot-exchange policy in Vietnam and find that consolidation raises land productivity. Kafle and Balasubramanya (2022) report for Niger that irrigation access is associated with about a 10 percent reduction in food insecurity and a 9 percent increase in food-consumption expenditure. Fischer et al. (2022) add an important caveat from semi-arid southern India: groundwater irrigation reduced overall poverty but increased the socioeconomic vulnerability of poorer households as aquifers became depleted, so the benefits of irrigation can be uneven and unsustainable if poorly managed. Mengistu and Kassie

(2022) and Hlatshwayo et al. (2022) find for Ethiopia and South Africa respectively that livestock and asset ownership, non-farm enterprise, and market participation reduce food insecurity, whereas drought shocks and high dependency ratios increase it.

T. Q. Tran and Vu (2021) find for about 2,500 households in Vietnam's North Central Coast that more fragmented cropland increases the likelihood of within-year hunger and the number of days without protein-rich food, with the burden concentrated among ethnic-minority households. Tu et al. (2021) find in the Vietnamese Mekong Delta that land accumulation improves both the technical and the environmental efficiency of rice production. Jambo et al. (2021) use propensity-score matching for Ethiopia and find that small-scale irrigation participation raises household income, food expenditure, and food security. Vu et al. (2021) show for rural India that larger landholdings and non-farm livelihood diversification are associated with better food and nutrition security.

Cholo et al. (2020) show for the Gamo Highlands of Ethiopia that fragmentation conditions the effect of land-management practices on technical efficiency and produces persistent yield gaps across a farmer's separate plots. Nguyen and Warr (2020) treat consolidation in Vietnam as a form of technical change and find that it raises farm productivity and income and promotes mechanisation while releasing labour to off-farm activities.

Ali et al. (2019) use panel data and a terrain-adjusted travel-cost measure of fragmentation in Rwanda. They confirm that fragmentation reduces yields, labour intensity, and technical efficiency, but they estimate the potential yield gain from consolidation at only about 7 percent and caution that, in an unmechanised setting, the costs of consolidation programmes may outweigh the benefits. This qualification, that the size of the fragmentation penalty depends on mechanisation and local production conditions, is central to the expectations of the present study.

Manjunatha et al. (2013), using stochastic frontier models on groundwater-irrigated farms in South India, find that land fragmentation is positively associated with farm inefficiency, whereas land ownership and crop diversity reduce it. The classic regional reference point remains Niroula and Thapa (2005), whose review of land fragmentation and consolidation across South Asia concludes that fragmentation, though occasionally useful for spreading agro-climatic risk, generally lowers efficiency by raising travel time,

boundary losses, and the costs of supervision and water management.

Taken together, this literature carries three lessons for the present study. First, districts with greater irrigation coverage should report lower production insufficiency, with magnitudes that depend on context. Second, reducing fragmentation can raise productivity, but largely through complementary inputs such as mechanisation and irrigation. Because fragmentation raises the costs of supervising scattered plots and of managing water across them (Niroula & Thapa, 2005), the disadvantage associated with fragmentation may be weaker where irrigation coverage is already high, or stronger if scattered plots make coordinated water use harder. None of the studies reviewed here estimates such a fragmentation-irrigation interaction directly, which motivates the explicit interaction term in this thesis. Third, food security is shaped by a combination of structural endowments, particularly land and water, and household access to assets and markets, which motivates the inclusion of control variables in the present analysis. Among these studies, Zheng et al. (2023), Phan et al. (2022), and T. Q. Tran and Vu (2021) provide the most direct international precedents for the outcome studied in this thesis.

2.2.2.2 National Context

Within Nepal, a growing body of empirical research has examined land fragmentation, irrigation, and food security, though usually as separate themes and at the local or household level. The studies are again presented from the most recent year to the earliest.

Paudel et al. (2025) describe a nexus of farmland abandonment, out-migration, and wildlife encroachment in the Karnali basin and show how depopulation and structural change reinforce the underuse of fragmented land.

Pun et al. (2024), using a randomised conjoint experiment, find that about 65 percent of Nepalese farmers prefer a farmland-consolidation policy, particularly when it is bundled with connection roads and irrigation. Nepal et al. (2024) assess three irrigation systems through a survey of 449 households and find that the most binding constraints are not physical infrastructure alone but timely access to fertiliser, the availability of source water, and fair output prices; they call for integrated rather than purely engineering solutions.

Shrestha et al. (2023) examine solar irrigation pumps in the Terai and caution that technology adoption is shaped by gender and social relations and may bypass women and marginal smallholders unless interventions are deliberately targeted.

Choudhary et al. (2022) estimate that Nepalese paddy farmers achieve only about 76 percent of potential output and identify land fragmentation, together with labour and capital constraints, as a key source of inefficiency, with small and marginal farms performing worse than larger ones. C. Dhakal and Escalante (2022) find that adopting improved organic-manure practices raised crop productivity by about 17 percentage points.

K. Khanal et al. (2021) estimate that the presence of a monetary fine raises the probability of participation in collective irrigation management by about 24 percentage points. Adhikari et al. (2021) find a mean technical efficiency of about 81 percent among wheat farmers and show that irrigation, improved varieties, and farmyard manure significantly raise efficiency. Pandey et al. (2021) show that male out-migration and the resulting feminisation of agriculture have ambiguous effects on food security: remittances ease food access in the short run, but domestic production suffers and land is left underused in the longer run.

Kumar et al. (2020) show with three rounds of the Nepal Living Standards Survey that dietary diversity improved over time largely because of changes in household and agricultural characteristics. Baral et al. (2020) document imbalanced and below-recommendation fertiliser application linked to weak extension services.

Kim et al. (2019) reach the same conclusion as the later migration work: male out-migration eases food access through remittances while threatening domestic production and contributing to land underuse.

B. N. Dhakal and Khanal (2018), in a household survey of Nawalparasi District, attribute fragmentation primarily to inheritance practices, population growth, active land markets, and urbanisation, and document that it lowers land productivity by increasing labour time, raising transport and boundary costs, and obstructing the use of chemical fertiliser and machinery. U. Khanal (2018) finds for Kaski District that about 40 percent of households kept land fallow and roughly 28 percent of plots were left uncultivated, even though sampled households produced enough food for only about seven months of the year. Fallowing decisions in that study were shaped by the number, size, and dis-

tance of plots, by irrigation status, and by out-migration; this is household-level evidence that parcel structure and water access jointly condition land use and, ultimately, sufficiency. T. R. Dhakal et al. (2018) identify the socio-economic characteristics of users and rule enforcement as drivers of collective action in farmer-managed irrigation systems (FMIS), which cover a large share of Nepal's irrigated area. Pariyar et al. (2018) show that access to water within mid-hill irrigation systems is unequally distributed across socio-economic groups and that equitable, farmer-controlled water allocation is key to raising productivity and reducing poverty. Merrey et al. (2018) document the fragile, climate-exposed livelihoods of high-altitude households in Rasuwa District.

Gartaula et al. (2017) describe a paradox in which food security improves through off-farm income even as the sustainability of small-scale agriculture declines. Joshi and Joshi (2017), applying a Tobit model to data from the previous agriculture census for eastern Nepal, find that landholding size, nearness to market, education, and engagement in agriculture significantly improve food security, while household size reduces it, and that hill and mountain households are more food insecure than those in the Terai. Morioka and Kondo (2017) find that agricultural productivity growth improved household food security even where holdings remained small and fragmented, with the largest gains among lower-income farmers. Ojha et al. (2017) argue that agricultural land in the Nepalese hills is increasingly underused through "socio-environmental pathways" driven by out-migration, declining farm profitability, and policy neglect.

Karki et al. (2015) report for the mid-hills that food self-sufficiency depends more on agronomic management (crop yields and improved cultivars) than on landholding size.

Three points from this national literature matter for the present study. First, while the irrigation studies establish who obtains water and how systems are governed, none of them quantifies the association between irrigation access and household production sufficiency in Nepal; the international estimates reviewed above therefore remain untested in the Nepali context. Second, the food-security literature leaves its central question unresolved: is sufficiency driven mainly by the structure of landholdings, as Joshi and Joshi (2017) find, or by how the land is managed, as Karki et al. (2015) report and as Kumar et al. (2020) and Morioka and Kondo (2017) also suggest? The two positions disagree on precisely the structural question this thesis examines, and nationwide evidence on land

structure and water access offers one way to adjudicate between them. Third, the migration studies caution that self-reported production sufficiency reflects farm structure and water access alongside livelihood changes that district-level controls capture only partially. The efficiency and input studies add that fragmentation and limited irrigation constrain efficiency, whereas irrigation and improved inputs help close yield gaps.

2.3 Policy Review

Land fragmentation and irrigation both occupy prominent places in Nepal’s agricultural policy architecture. The Agriculture Development Strategy 2015–2035, the apex sectoral strategy, identifies small and fragmenting holdings as a constraint on commercialisation, lists land consolidation among its policy instruments, and sets explicit targets for expanding year-round irrigation (MoAD, 2015). Successive editions of the Economic Survey likewise treat irrigation expansion and land productivity as central policy concerns while acknowledging that a majority of agricultural holdings still do not produce enough for their own consumption (MoF, 2024). Since federal restructuring, agricultural extension and most local irrigation functions rest with provincial and local governments, which makes district-level evidence on where structural disadvantage is concentrated directly usable for sub-national planning.

Two more recent instruments bear directly on the fragmentation side of this study. The National Land Policy 2019 commits the state to promoting land consolidation and discouraging the further subdivision of agricultural land, alongside its tenure-security and land-administration agendas (MoLMCPA, 2019). The Land Use Act 2019 classifies all land into use categories, of which agriculture is one, restricts the conversion of agricultural land to other uses and the construction of new structures on it, and creates federal, provincial, and local land-use councils to enforce the classification (GoN, 2019). Both instruments presume that keeping consolidated agricultural land in agricultural use raises productivity and food security; the district-level evidence of this thesis speaks to exactly that presumption.

Two features of this policy context matter for the present study. First, consolidation and irrigation are usually pursued through separate policy instruments, while the literature reviewed above suggests that their effects are intertwined; evidence on their joint

distribution and interaction therefore speaks to the design of both. Second, policy targets are formulated at the national level, whereas the binding constraints documented in the literature are local and heterogeneous; district-level evidence addresses exactly that mismatch.

2.4 Research Gap

The reviewed literature establishes two robust regularities. Internationally, land fragmentation is generally associated with lower technical and economic efficiency and, in the most recent studies, with greater household food insecurity, while irrigation is associated with higher productivity, income, and food security. However, this evidence is overwhelmingly based on household- or plot-level data from other countries, and it rarely studies fragmentation and irrigation jointly as correlates of a household production-sufficiency outcome.

The nearest Nepalese precedents fall short of this combination in identifiable ways. U. Khanal (2018) models plot numbers, plot distance, irrigation status, and migration jointly, but for fallowing decisions in a single district. Joshi and Joshi (2017) use the previous agriculture census of 2011/12 to study food-security determinants, but only for the eastern region and without a fragmentation measure. Choudhary et al. (2022) link fragmentation and input constraints to farm efficiency, but in two districts and with an efficiency rather than a sufficiency outcome. No existing study provides nationwide, district-level evidence that examines land fragmentation and irrigation access *together*, including their interaction, and links them to agricultural production sufficiency. The district tables of the NSCA 2021/22, which report consistent information on parcel structure, irrigation, sufficiency, and relevant controls for all 77 districts, have not yet been combined into a single reproducible empirical framework for this purpose.

This study addresses these gaps in three ways. First, it uses the most recent nationwide agriculture census to construct a consistent district-level dataset covering all 77 districts. Second, it analyses land fragmentation and irrigation access jointly, including their interaction, rather than in isolation. Third, it focuses on agricultural production sufficiency (whether holdings produce enough for household consumption) as a direct and policy-relevant welfare outcome. In doing so, the study complements the household-

level international evidence with district-level evidence from Nepal and establishes a transparent, reproducible workflow that can be extended when household microdata become available.

In terms of the objectives stated in Section 1.3, the first and second objectives address the first gap: the absence of nationwide descriptive evidence on how fragmentation, irrigation, and sufficiency are distributed across districts. The third objective addresses the second gap: the absence of multivariate district-level evidence linking the two structural conditions and their interaction to sufficiency. The methodological gap is addressed by the workflow itself: because every estimate derives from published official tables through documented scripts, the framework can be re-estimated directly when the next census round or comparable district data become available.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes how the study is designed and carried out. It first explains the research design and the variables studied, then develops the theoretical and conceptual framework that links land fragmentation and irrigation access to agricultural production sufficiency, describes the nature and sources of the data and the study area, and then sets out the empirical models and the operational definitions of all variables.

3.2 Research Design

The study follows a quantitative, cross-sectional, district-level research design. It is descriptive in its first stage, documenting the spatial patterns of fragmentation, irrigation, and production sufficiency across all districts of Nepal, and analytical in its second stage, estimating multivariate associations between these conditions with appropriate controls and fixed effects. The design is appropriate because the research questions concern broad structural variation across districts rather than individual farm behaviour, and because the only nationwide data source that reports all the required variables in a consistent format, the National Sample Census of Agriculture (NSCA) 2021/22, is published at the district level. All results are interpreted as district-level associations rather than household-level causal effects.

The reliance on secondary census data is itself a design choice with clear advantages for the questions asked. The NSCA is collected by the National Statistics Office under a uniform national methodology, so its district figures are consistent and comparable in a way that no feasible primary survey by a single researcher could match; its coverage of all 77 districts eliminates site-selection bias; and its public availability makes every step of the analysis verifiable. The corresponding costs, the fixed set of variables and the district resolution, are acknowledged in Section 1.6 and shape the interpretation of the results throughout.

The dependent variable of the study is the agricultural production insufficiency share: the share of holdings in a district reporting that their own agricultural production is insufficient for household consumption, the complement of the census sufficiency measure. The principal independent variables are land fragmentation, measured as the average number of parcels per holding, and irrigation access, measured as the irrigated share of the holding area, together with their interaction. The specification is completed by six district-level control variables and by province (or, in robustness checks, ecological-belt) fixed effects; the controls are average holding size, the agricultural-loan share, the government-subsidy share, the mechanisation (power-tiller) share, the fast-market-access share, and the climate-impact share. Section 3.9 defines every variable operationally, and Section 3.7 states the estimating equation.

3.3 Theoretical and Conceptual Framework

The analysis rests on the standard farm-household production framework, in which a holding combines land, labour, water, and purchased inputs to produce output that is either consumed by the household or sold. Whether the holding's own production is sufficient for household consumption then depends on the productivity of its resources relative to household needs. Within this framework, the two structural conditions studied here operate through distinct mechanisms.

Land fragmentation affects the production possibilities of a holding primarily through transaction and supervision costs. Scattered parcels increase travel time between plots, raise the cost of supervising labour and moving equipment, generate boundary losses, and complicate the application of water, fertiliser, and machinery (B. N. Dhakal & Khanal, 2018; Niroula & Thapa, 2005). Against these costs, the literature recognises a risk-diversification benefit: parcels spread over different micro-environments can insure the household against localised weather, pest, or soil shocks. The net effect of fragmentation is therefore theoretically ambiguous and ultimately an empirical question, although the dominant expectation for a labour-constrained agriculture such as Nepal's is negative (Hao et al., 2023; Phan et al., 2022).

Irrigation enters the framework as a risk-reducing and productivity-enhancing input. Reliable water access weakens the dependence of production on the monsoon, permits

multiple cropping cycles, and raises the return to complementary inputs such as improved seeds and fertiliser (Jambo et al., 2021; Mupaso et al., 2023). Districts where a larger share of agricultural land is irrigated are therefore expected to convert their land endowment into household food needs more reliably than rainfed districts.

The interaction between the two conditions completes the framework, and its sign depends on which margin of the fragmentation penalty dominates. Along the *water-management* margin, part of the fragmentation penalty arises because scattered plots are harder to irrigate in a coordinated way; assured irrigation then substitutes for the coordination that fragmentation frustrates, so the two are substitutes and the interaction term is expected to be negative (irrigation softens the fragmentation disadvantage). Along the *labour-supervision* margin, however, fragmentation chiefly raises travel and supervision costs and impedes the mechanisation that irrigation makes worthwhile; if irrigation only pays off on consolidated, mechanisable plots, the two are complements and the interaction is expected to be positive. Because the framework admits both margins, the sign of the interaction is left as an empirical question, and the estimate is interpreted as a test of which margin dominates rather than merely as the presence or absence of moderation.

Two identification threats follow directly from the framework and discipline the interpretation of all results. First, irrigation placement is not random: surface irrigation concentrates in the Terai plains for reasons of terrain and historical canal investment, while production insufficiency concentrates in the mountains. Any district-level association between irrigation and sufficiency therefore risks absorbing broad agro-ecological differences. The empirical strategy responds with province fixed effects in the main models and with ecological-belt (Mountain/Hill/Terai) controls in the robustness analysis. The fixed effects absorb some, but not all, of this confound: historical canal investment tracks the same Terai fertility and terrain that independently raise sufficiency, and public irrigation programmes may themselves respond to observed deprivation, so selection and reverse causality cannot be excluded within a single cross-section. An ideal design would instrument irrigation access with a determinant of feasibility that does not otherwise affect sufficiency, such as slope-based gravity-fed feasibility or proximity to canal command areas; constructing such an instrument requires geographic data beyond the census and is beyond the scope of this thesis, which is one more reason the estimates

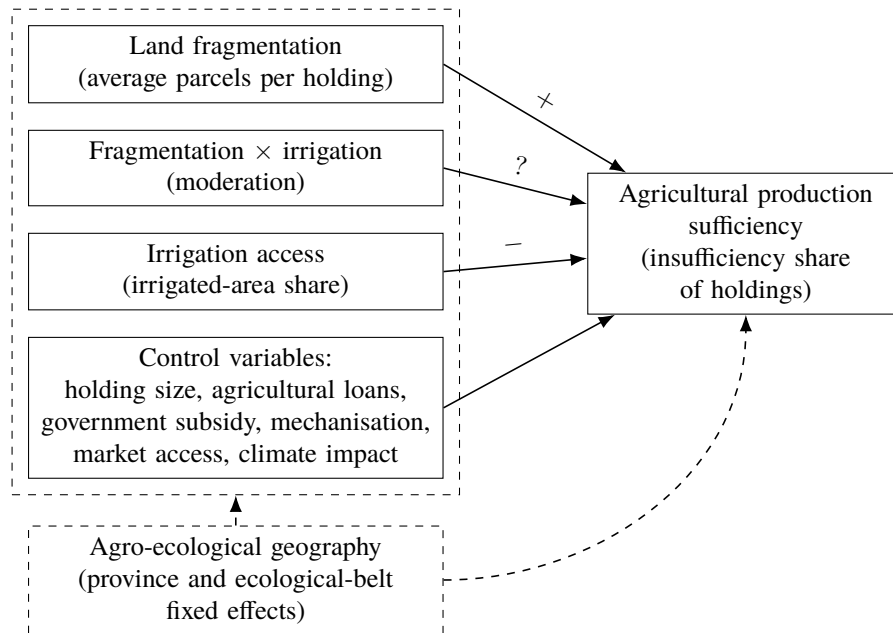
are read throughout as conditional associations. Second, the outcome is the holder's self-reported assessment of sufficiency, which measures production relative to household needs rather than poverty or food security; this shapes the operational definitions in Section 3.9 and the interpretation throughout Chapters IV and V.

Two consequences of the district unit of analysis also follow from the framework. Because the regression relates district aggregates rather than individual holdings, its coefficients are subject to the ecological-inference problem: associations estimated across areal units need not equal the corresponding household-level relationships, and they are sensitive to how the areal units are drawn: the modifiable areal unit problem (King, 1997; Openshaw, 1984). The study therefore confines its claims to the district level throughout. In addition, the household production framework implies an omitted-variable concern that the available data cannot fully close. Remittance income, which the migration literature reviewed in Section 2.2.3 identifies as central in rural Nepal (Kim et al., 2019; Pandey et al., 2021), is plausibly correlated both with fragmentation (out-migration leaves land underutilised and finances its subdivision through inheritance) and with the outcome (remittances fund food purchases, decoupling sufficiency from own production). Its omission, and that of average household size, would tend to bias the fragmentation coefficient away from zero and to load part of the sufficiency variation onto the included structural variables; the estimated structural associations are read with this in mind.

Figure 3.1 summarises the framework as a block diagram: the two structural conditions and their interaction, together with the district controls, are related to production sufficiency, while agro-ecological geography, absorbed by the fixed effects, shapes both the structural conditions and the outcome.

The three hypotheses implied by this framework are stated in Section 1.4 and are tested through the empirical model specified in Section 3.7. Each concerns a conditional district-level association, given the controls and the fixed effects, rather than a household-level causal effect, so the identification threats discussed above apply to all three.

Figure 3.1
Conceptual Framework of the Study



Note: solid arrows show the hypothesised district-level associations, with expected signs in parentheses along each arrow. Dashed arrows indicate that agro-ecological geography shapes both the structural conditions and the outcome; the province and ecological-belt fixed effects absorb this common influence.

3.4 Nature and Sources of Data

The study is based entirely on secondary data. The principal source is the NSCA 2021/22, published by the National Statistics Office (NSO, 2023), whose district tables provide information on parcel structure, irrigation, agricultural production sufficiency, the duration of food insufficiency, agricultural loans, government subsidy, agricultural implements, climate impact, and access to agricultural markets. The Economic Survey and the Agriculture Development Strategy provide background and policy context (MoAD, 2015; MoF, 2024). The relevant district tables were obtained in machine-readable form from the National Statistics Office's open-data portal and were cleaned and merged by district into a single research dataset using documented Python scripts, so that every step from the official tables to the final estimates is reproducible; the merged dataset and its mapping to the official census tables are reported in Annex I. Concretely, each census table was first reduced to a district-level series; the series were then merged on the district name, with spellings harmonised to the 77-district federal structure; every share was computed from the underlying counts and areas rather than transcribed; and the resulting dataset was verified to cover all 77 districts with no

missing values in any variable used in the models.

3.4.1 The National Sample Census of Agriculture 2021/22

The NSCA 2021/22 is a sample census of agriculture: rather than enumerating every holding in the country, it estimates national, provincial, and district figures from a large probability sample of agricultural holdings (NSO, 2023). Its primary reference period is the agricultural year 2021/22, and the main fieldwork was carried out over a forty-five-day period between 19 April and 2 June 2022 (NSO, 2023). The census was conducted by the National Statistics Office—the successor to the former Central Bureau of Statistics—and covers all seventy-seven districts, seven provinces, and every local level of the federal structure (NSO, 2023).

The sampling frame was built from the enumeration areas delineated for the 2021 National Population and Housing Census, together with the holding information collected in those areas (NSO, 2023). The design is a stratified two-stage sample. At the first stage, enumeration areas are the primary sampling units and are selected by probability proportional to size, with the number of holdings in each area serving as the measure of size; at the second stage, agricultural holdings within each selected area are the units of enumeration (NSO, 2023). Within each municipality, holdings are implicitly stratified by size class before selection, so that small, medium, and large holdings—as well as livestock-only holdings—are each represented, and the seven hundred fifty-three local levels serve as the domains of estimation (NSO, 2023). In total, about 13,576 enumeration areas were selected and some 330,112 holdings were enumerated nationwide, representing approximately eight percent of all agricultural holdings in the country (NSO, 2023).

3.4.2 Data Collection and the Mapping of Variables

Data collection used a set of complementary instruments (NSO, 2023). A holding listing form was first used to enumerate all agricultural holdings above the census cut-off within each selected enumeration area and to serve as the frame from which holdings were drawn; a short selected-holding form recorded the holdings chosen for detailed

interview; the main agriculture holding questionnaire then collected the substantive holding-level information; and a community questionnaire, administered at the ward office, captured community-level characteristics (NSO, 2023). The main questionnaire records the characteristics of the holding and its operator, the number and area of parcels and their land use, the source and extent of irrigation, crops and livestock, the ownership of agricultural implements and machinery, the use of agricultural loans and government subsidy, the sufficiency of agricultural production and the number of months of any food insufficiency, the impact of climate and related hazards, and access to agricultural markets (NSO, 2023). Every variable used in this study is a holdings-based district aggregate of one of these questionnaire items, as set out in the operational definitions of Section 3.9.

A further point about this sample design matters for inference. Because the district figures are estimated from a probability sample rather than a complete enumeration, every district share used in this study is itself an estimate subject to sampling error. Although the study covers all 77 districts (the complete population of districts), the reported standard errors remain meaningful under model-based, super-population inference: the district observations are treated as draws from a data-generating process whose parameters are the objects of interest, and the sampling variability embedded in the census estimates provides a further reason not to treat the observed shares as exact population constants.

3.5 Study Area

The study area is the whole of Nepal at district resolution: all 77 districts across the seven provinces and the three ecological belts. Sixteen districts are classified as Mountain, 40 as Hill, and 21 as Terai under the standard classification, and the belt of each district is reported in Annex I. The districts span the full range of Nepalese agricultural conditions, from high-altitude subsistence systems with almost no irrigation (Rukum East, 2.9 percent of area irrigated) to intensively irrigated Terai districts (Bara, 91.1 percent). Because the study uses published census tables rather than a primary survey, no sample-size calculation is required: the analysis covers the complete population of 77 districts, the underlying holding-level information having been collected by the National Statistics Office through the probability sampling design of the NSCA 2021/22. The unit of

analysis is the district, and the dataset contains no missing values for any variable used in the models.

3.6 Tools of Analysis and Empirical Model

The analysis proceeds in three stages. First, descriptive statistics, province profiles, district rankings, and figures establish the spatial patterns of the main variables. Second, simple and holdings-weighted correlations summarise the bivariate relationships with the outcome. Third, multivariate models estimate the conditional associations of interest.

Two estimators are used. A weighted ordinary least squares (OLS) regression provides a transparent benchmark whose coefficients are directly interpretable in percentage points. Because the dependent variable is a proportion bounded between zero and one, the preferred specification is a fractional logit estimated by quasi-maximum likelihood (Papke & Wooldridge, 1996), which respects the bounded support and yields fitted values inside the unit interval by construction. Robust (heteroskedasticity-consistent) standard errors (SE) are reported for both estimators, and statistical significance is assessed with the large-sample normal approximation. Results from the fractional logit are translated into interpretable quantities through average partial effects (APE) and scenario predictions. Formally, with w_d the district analytic weight (total holdings, normalised to mean one) and x_d the regressor vector, the weighted OLS estimator minimises $\sum_d w_d (y_d - x'_d \beta)^2$, and its coefficient covariance is estimated with the HC1 sandwich

$$\widehat{V}(\hat{\beta}) = \frac{n}{n-k} \left(\sum_d w_d x_d x'_d \right)^{-1} \left(\sum_d w_d^2 \hat{\varepsilon}_d^2 x_d x'_d \right) \left(\sum_d w_d x_d x'_d \right)^{-1}, \quad (3.1)$$

where $n = 77$ and k is the number of estimated parameters. The fractional logit maximises the weighted Bernoulli quasi-log-likelihood

$$\ell(\beta) = \sum_d w_d \left[y_d \ln \Lambda(x'_d \beta) + (1 - y_d) \ln \{1 - \Lambda(x'_d \beta)\} \right], \quad (3.2)$$

where $\Lambda(\cdot)$ is the logistic function. As Papke & Wooldridge (1996) show, the resulting estimator is consistent whenever the conditional mean $E[y_d | x_d] = \Lambda(x'_d \beta)$ is correctly

specified, even though y_d is a share rather than a binary outcome; the equation is solved by Newton iteration and inference again uses a robust sandwich covariance built from the weighted scores. Multicollinearity among the regressors is monitored with variance inflation factors, $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is the coefficient of determination from regressing regressor j on all the other regressors.

Three further methodological choices respond to the small-sample, district-level character of the data. First, the main models weight districts by their total number of agricultural holdings, normalised so that inference remains district-level; the estimand is thus the association experienced by the average agricultural holding rather than the average district. Because weighting can change estimates when associations are heterogeneous, unweighted estimates are reported as a robustness check (Solon et al., 2015). Second, the full specification estimates 16 parameters on 77 observations (fewer than five observations per parameter), so a parsimonious specification retaining only fragmentation, irrigation, their interaction, holding size, and province fixed effects is estimated alongside it, and conclusions rest on the stability of signs and magnitudes across specifications rather than on any single p-value. Third, a pairs (district) bootstrap with 999 replications provides distribution-free standard errors and percentile confidence intervals for the irrigation coefficient and for the scenario gap between low- and high-irrigation profiles. The bootstrap proceeds as follows: (i) draw 77 districts with replacement from the sample, keeping each drawn district's outcome, regressors, and weight; (ii) re-estimate the main fractional logit on the resampled data; (iii) record the irrigation coefficient and the two scenario gaps; (iv) repeat 999 times; and (v) report the standard deviation of each recorded quantity as its bootstrap standard error, and the 2.5th and 97.5th percentiles of its bootstrap distribution as the 95 percent confidence interval.

Two further inferential issues are acknowledged rather than fully solved, in keeping with the district-level design. Districts within a province share policies, extension services, and weather shocks, so the disturbances are plausibly correlated within province; the natural response, clustering the standard errors by province, is unreliable here because there are only seven provinces, far below the number of clusters the cluster-robust asymptotics require. The heteroskedasticity-robust standard errors are retained for that reason, and the province fixed effects absorb the province-common component of the disturbance directly; a wild cluster bootstrap, designed precisely for settings with few clusters

(Cameron et al., 2008), is a refinement left for future work. Relatedly, the district observations are spatial units and the key variables are spatially structured, so residual spatial autocorrelation is a first-order concern: positive autocorrelation would make the robust standard errors optimistic. The study therefore tests for it directly. A queen-contiguity matrix for the 77 districts is constructed from the official district boundaries, and Moran's I is computed on the residuals of both main models with permutation-based inference; the results, reported in Section 4.4.3, show no evidence of positive spatial autocorrelation, indicating that the province fixed effects absorb the spatial structure of the data. A full spatial-error or spatial-lag model is nevertheless left for future work, and the reported standard errors are read as indicative rather than exact.

3.7 Model Specification

The main estimating equation is

$$\begin{aligned}
Y_d = & \beta_0 + \beta_1 Frag_d + \beta_2 Irrig_d + \beta_3 (Frag_d \times Irrig_d) \\
& + \beta_4 Size_d + \beta_5 Loan_d + \beta_6 Subsidy_d + \beta_7 Mech_d \\
& + \beta_8 Market_d + \beta_9 Climate_d + \delta_p + \varepsilon_d,
\end{aligned} \tag{3.3}$$

where Y_d is the agricultural production insufficiency share of district d , $Frag_d$ is the average number of parcels per holding, $Irrig_d$ is the irrigated share of the holding area, δ_p denotes province fixed effects, and the remaining regressors are the district controls defined in Section 3.9. In the fractional logit the same index enters through the logistic conditional mean, $E[Y_d | X_d] = \Lambda(X_d' \beta)$, estimated by Bernoulli quasi-maximum likelihood (Papke & Wooldridge, 1996).

In terms of these coefficients, the three hypotheses of Section 1.4 are stated formally as H1: $\beta_2 < 0$ (wider irrigation coverage is associated with lower insufficiency); H2: $\beta_1 > 0$ (greater fragmentation is associated with higher insufficiency); and H3: $\beta_3 \neq 0$ (irrigation moderates the fragmentation association), where the water-management margin implies $\beta_3 < 0$ and the labour-supervision margin implies $\beta_3 > 0$.

A specification with an interaction term raises a multicollinearity concern, because the interaction is by construction highly correlated with its two components. The study deals

with this in the standard way: fragmentation and irrigation are centred at their holdings-weighted means (2.81 parcels per holding and 0.525 irrigated-area share) before the interaction is formed, so that the interaction regressor is $(Frag_d - \overline{Frag})(Irrig_d - \overline{Irrig})$. Mean-centring leaves the model's fit, its predictions, and the interaction coefficient itself unchanged, but it makes β_1 and β_2 directly interpretable as the associations evaluated at the average holding's profile, rather than at the out-of-sample point where the other variable equals zero, and it removes the mechanical collinearity that would otherwise inflate their standard errors. Multicollinearity is monitored with variance inflation factors (VIF): after centring, every VIF in the main specification lies below the conventional threshold of 10 (Table 4.10), whereas without centring the VIFs of irrigation and of the interaction would be 17.7 and 13.9. The robustness analysis re-estimates Equation (3.3) with ecological-belt fixed effects in place of (and in addition to) province effects, with the irrigated-holdings share replacing the irrigated-area share, with the complementary sufficiency share and the severe 10–12-month food-insufficiency share as alternative outcomes, with an average-household-size control and a parcels-per-hectare fragmentation proxy, and in the parsimonious and unweighted variants described above.

3.8 Definition of Key Terms

For clarity, the key terms of the study are used in the following senses throughout the thesis; Section 3.9 gives the exact operational construction of every variable.

Land fragmentation refers to the division of a single agricultural holding into several spatially separated parcels. The study measures it by the average number of parcels per holding in a district.

Irrigation access refers to the availability of water for cultivation beyond rainfall. The study measures it by the share of a district's total holding area that is irrigated, with the share of holdings reporting any irrigation used as an alternative measure.

Agricultural production sufficiency refers to whether a holding's own agricultural production is sufficient for its household's consumption, as reported by the holder in the census. The analysis models its complement, the **insufficiency share**: the proportion of holdings in a district reporting insufficient own production.

Severe food insufficiency refers to holdings reporting that food from own production and other sources was insufficient for 10 to 12 months of the year. It is a duration-based census measure, distinct from production sufficiency, and is treated as a separate dimension of deprivation.

Ecological belt refers to the standard division of Nepal’s districts into the Mountain, Hill, and Terai belts, which summarises the country’s agro-ecological geography. **Province** refers to the seven federal provinces. The **district**, of which there are 77, is the unit of analysis of the study.

3.9 Operational Definition of the Variables

Table 3.1 defines every variable, its construction, its source table in the NSCA 2021/22, and its expected association with the insufficiency outcome. Two definitional points deserve emphasis. First, the outcome derives from NSCA Table 35, in which the holder reports whether the holding’s own agricultural products were *sufficient* for household consumption; it is a self-reported, production-relative-to-needs measure, and its complement (the insufficiency share) is the modelled outcome. Second, the severe measure derives from NSCA Table 36.2, which records the *duration* of food insufficiency; holdings reporting 10 to 12 months are classified as severely insufficient. The two census constructs measure different dimensions of deprivation and are treated as distinct outcomes throughout.

Justification of the Variable Selection

The variables are not chosen ad hoc: each follows from the farm-household framework of Section 3.3, corresponds to a determinant already established in the literature of Chapter II, and maps to a specific NSCA 2021/22 district table (Table 3.1). The two structural regressors carry the study’s hypotheses. Average parcels per holding (*Frag_d*, Table 3) operationalises the transaction- and supervision-cost channel of fragmentation (B. N. Dhakal & Khanal, 2018; Niroula & Thapa, 2005), whose penalty the international evidence finds to run mainly through impeded machinery use and labour productivity (Hao et al., 2023); the irrigated-area share (*Irrig_d*, Table 4.1) operationalises irrigation as

the risk-reducing, productivity-enhancing input of the household model (Jambo et al., 2021; Mupaso et al., 2023). Their product ($Frag_d \times Irrig_d$) is included because no study reviewed above estimates this moderation directly, yet the same literature implies that scattered plots are harder to irrigate and mechanise in a coordinated way (Niroula & Thapa, 2005).

The controls hold constant the remaining determinants of production sufficiency that Chapter II identifies, so that the structural associations are not confounded by them. Average holding size ($Size_d$, Table 3) is the census analogue of the farm-size variable that Joshi and Joshi (2017) and Phan et al. (2022) find protective of food security. The agricultural-loan share ($Loan_d$, Table 17) proxies credit access, which Phan et al. (2022) identify as reducing severe food insecurity, and the government-subsidy share ($Subsidy_d$, Table 19) proxies the public input support whose absence the national extension literature links to below-optimal input use (Baral et al., 2020). The power-tiller share ($Mech_d$, Table 14) proxies mechanisation, the very margin through which fragmentation is found to bite hardest (B. N. Dhakal & Khanal, 2018; Hao et al., 2023), and is therefore both a determinant of sufficiency in its own right and the channel the interaction is meant to probe. The 30-minute market-access share ($Market_d$, Table 38.1) captures the market integration that Joshi and Joshi (2017) and Mengistu & Kassie (2022) find lowers food insecurity, and the climate-impact share ($Climate_d$, Table 22) captures self-reported exposure to the adverse-weather shocks that raise it (Mengistu & Kassie, 2022; Merrey et al., 2018) and that irrigation is expected to buffer. Every control is expressed as a holdings-based district share, so all regressors enter the models on a comparable scale. Determinants that the framework flags but the census cannot measure—chiefly remittance income and household size—are left to the omitted-variable discussion of Section 3.3 and to a household-size robustness check.

Table 3.1
Operational Definitions of the Variables

Variable	Category	Construction and interpretation	NSCA source	Expected sign
Y_d	Dependent	Holdings reporting own agricultural products insufficient for household consumption, divided by total holdings (self-reported)	Table 35	—
$Severe_d$	Dependent	Holdings reporting food insufficiency for 10–12 months, divided by total holdings	Table 36.2	—
$Frag_d$	Independent	Total parcels divided by total holdings (average parcels per holding)	Table 3	+
$Irrig_d$	Independent	Irrigated area divided by total area of holdings	Table 4.1	—
$Frag_d \times Irrig_d$	Independent	Interaction of fragmentation and irrigation	Tables 3, 4.1	?
$Size_d$	Control	Total holding area divided by total holdings (ha)	Table 3	—
$Loan_d$	Control	Holdings with an agricultural loan, divided by total holdings	Table 17	—
$Subsidy_d$	Control	Holdings receiving government subsidy, divided by total holdings	Table 19	—
$Mech_d$	Control	Holdings using a power tiller, divided by total holdings	Table 14	—
$Market_d$	Control	Holdings within 30 minutes on foot of an agricultural market, divided by total holdings	Table 38.1	—
$Climate_d$	Control	Holdings reporting adverse climate impact on agriculture, divided by total holdings	Table 22	+

Table 3.1*Operational Definitions of the Variables (continued)*

Variable	Category	Construction and interpretation	NSCA source	Expected sign
Province, belt	Fixed effects	Province of the district; ecological belt (Mountain, Hill, or Terai)	—	—

CHAPTER IV

DATA PRESENTATION AND ANALYSIS

4.1 Introduction

This chapter presents the data and the empirical results. After a brief overview of the study, Section 4.3 documents the district-level patterns of fragmentation and irrigation (objective 1) and of agricultural production sufficiency (objective 2), Section 4.4 analyses the multivariate relationship between them, including the interaction, the robustness checks, and the severe food-insufficiency outcome (objective 3), Section 4.5 reports the regression diagnostic tests, and Section 4.6 discusses the findings against the literature reviewed in Chapter II.

4.2 Overview of the Study

The analysis covers all 77 districts of Nepal using the district research dataset constructed from the NSCA 2021/22 tables (Annex I). The outcome is the share of holdings reporting that their own agricultural production is insufficient for household consumption; the key explanatory variables are average parcels per holding and the irrigated-area share, together with their interaction and the controls defined in Table 3.1. Estimation follows the sequence of Chapter III: descriptive and correlation analysis, weighted OLS, the preferred fractional logit with province fixed effects, average partial effects and scenario predictions with bootstrap inference, and an extensive set of robustness specifications.

4.3 District-Level Patterns of Fragmentation, Irrigation, and Sufficiency

4.3.1 Descriptive Patterns

Table 4.1 reports the summary statistics. The picture is one of small, fragmented, and unevenly watered agriculture. The weighted mean holding operates only 0.537 ha divided into 2.81 parcels, and just over half of the holding area (52.5 percent) is irrigated;

this figure is the holdings-weighted mean of the district shares and therefore differs slightly from the census's national area aggregate of 54.5 percent cited in Chapter I. The weighted mean insufficiency share is 0.550: for the majority of Nepalese holdings, own production does not cover household consumption. Dispersion is wide on every dimension: parcels per holding range from 1.5 (Lalitpur) to 6.3 (Humla), the irrigated-area share from 0.029 (Rukum East) to 0.911 (Bara), and the insufficiency share from 0.319 (Rupandehi) to 0.869 (Dolpa).

Table 4.1
Summary Statistics of the District Variables

Variable	Mean	Weighted mean	Weighted SD	Min	Max
Average parcels per holding	3.101	2.806	0.787	1.500	6.300
Average holding size (ha)	0.509	0.537	0.141	0.146	0.844
Irrigation area share	0.421	0.525	0.258	0.029	0.911
Irrigated holdings share	0.639	0.699	0.179	0.112	0.970
Agricultural sufficiency share	0.387	0.450	0.152	0.131	0.681
Agricultural insufficiency share	0.613	0.550	0.152	0.319	0.869
Loan share	0.100	0.117	0.040	0.015	0.185
Subsidy share	0.076	0.075	0.078	0.005	0.560
Power tiller share	0.102	0.114	0.118	0.000	0.498
Climate impact share	0.428	0.397	0.094	0.207	0.840
Severe insufficiency share (10-12 months)	0.039	0.049	0.033	0.001	0.163
Fast market access share	0.156	0.151	0.060	0.057	0.410

Note: 77 districts. Weighted statistics use the total number of agricultural holdings in each district as weights. Source: author's computations from the NSCA 2021/22 district tables (NSO, 2023).

Table 4.2 aggregates the patterns to provinces and Figure 4.1 visualises them. The inverse association between irrigation and insufficiency is already visible at this resolution: Madhesh combines the highest weighted irrigation coverage (0.743) with the lowest insufficiency (0.430), while Karnali combines the lowest irrigation coverage (0.289) with the highest insufficiency (0.695), followed by Gandaki (0.650). Fragmentation is highest in Karnali (3.35 parcels per holding) and lowest in Koshi (2.20).

The remaining provinces fill in the gradient. Lumbini, with substantial Terai territory, combines above-average irrigation (0.561) with below-average insufficiency (0.510); Koshi shows a similar profile (0.542 irrigated, 0.559 insufficient) with the least fragmented holdings in the country. Bagmati is the instructive outlier: its irrigation coverage is low (0.393), yet its insufficiency (0.559) is no worse than Koshi's, reflecting the Kathmandu valley's market production and off-farm income, an anomaly that reappears as a significant province fixed effect in Section 4.4.1. Sudurpashchim, despite irrigation

coverage comparable to Koshi's (0.562), reports higher insufficiency (0.599), consistent with its heavily fragmented far-western hill districts (3.19 parcels per holding). Table 4.3 lists the district extremes: the five highest-insufficiency districts are all mountain or hill districts (Dolpa, Manang, Bajura, Arghakhanchi, Pyuthan), while the five lowest are all Terai districts.

Table 4.2

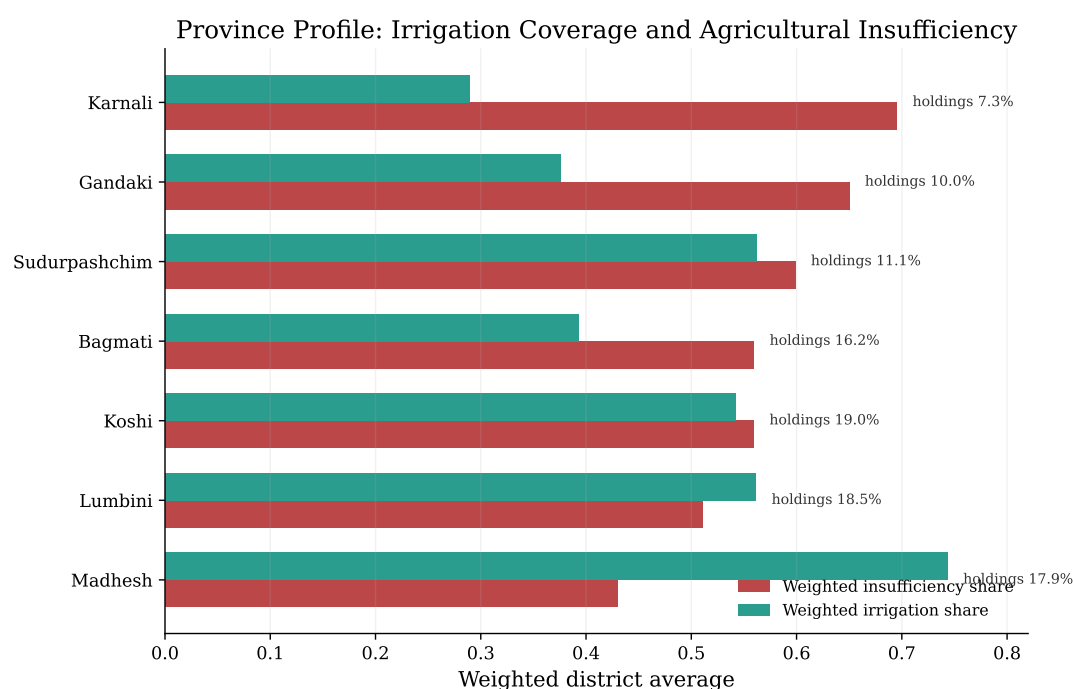
Province Profile of Fragmentation, Irrigation, and Insufficiency

Province	Districts	Holdings share	Parcels	Irrig. share	Insuff. share
Koshi	14	0.190	2.201	0.542	0.559
Madhesh	8	0.179	2.968	0.743	0.430
Bagmati	13	0.162	2.579	0.393	0.559
Gandaki	11	0.100	2.986	0.376	0.650
Lumbini	12	0.185	2.927	0.561	0.510
Karnali	10	0.073	3.353	0.289	0.695
Sudurpashchim	9	0.111	3.192	0.562	0.599

Note: weighted province means using district holdings as weights. Source: author's computations from the NSCA 2021/22 district tables.

Figure 4.1

Weighted Province Profile of Irrigation Coverage and Agricultural Insufficiency



Note: Province values are weighted by district agricultural holdings. Madhesh stands out for high irrigation and lower insufficiency, while Karnali shows the opposite pattern.

Table 4.3*Districts With the Highest and Lowest Agricultural Insufficiency Shares*

Rank	Highest insufficiency		Lowest insufficiency	
	District (province)	Share	District (province)	Share
1	Dolpa (Karnali)	0.869	Rupandehi (Lumbini)	0.319
2	Manang (Gandaki)	0.866	Bardiya (Lumbini)	0.346
3	Bajura (Sudurpashchim)	0.857	Parsa (Madhesh)	0.351
4	Arghakhanchi (Lumbini)	0.854	Nawalparasi West (Lumbini)	0.352
5	Pyuthan (Lumbini)	0.835	Kanchanpur (Sudurpashchim)	0.365

4.3.2 Variation in Agricultural Production Sufficiency

The second objective, and the second research question, concern how agricultural production sufficiency itself varies across the districts of Nepal. The outcome is summarised here as the insufficiency share, the complement of the census sufficiency measure. Production insufficiency is the national norm rather than the exception: the holdings-weighted mean insufficiency share is 0.550, so a majority of Nepalese holdings report that their own production does not cover household consumption. The variation around this mean is wide and clearly geographic; across districts the insufficiency share ranges almost threefold, from 0.319 in Rupandehi to 0.869 in Dolpa (Table 4.1).

The province profile in Table 4.2 and Figure 4.1 locates this variation. Weighted insufficiency is lowest in Madhesh (0.430) and highest in Karnali (0.695), followed by Gandaki (0.650), so the burden of insufficiency is concentrated in the mountain and hill provinces and is lightest in the well-irrigated Terai plains. The district extremes in Table 4.3 sharpen the same picture: the five highest-insufficiency districts are all mountain or hill districts (Dolpa, Manang, Bajura, Arghakhanchi, and Pyuthan), while the five lowest are all Terai districts. The ecological-belt gradient described in Section 4.3.3 confirms the pattern, with the insufficiency share falling from 0.711 in the Mountain belt to 0.422 in the Terai.

This descriptive answer to the second research question is the mirror image of the irrigation gradient documented above, which anticipates the multivariate analysis of Section 4.4. It should be read carefully, however, because this overall sufficiency measure has a different, and indeed nearly opposite, geography from the severe 10–12-month food-insufficiency measure examined in Section 4.4.4; the two are therefore treated through-

out as distinct dimensions of deprivation.

4.3.3 Patterns by Ecological Belt

Because the robustness analysis of Section 4.4.3 relies on ecological-belt controls, the belt-level pattern deserves its own description. Table 4.4 aggregates the three core variables and the severe share to the Mountain, Hill, and Terai belts, and Figure A3 in Annex III shows the underlying district distribution of irrigation by belt. The gradients are stark. The 16 Mountain districts hold only 8.6 percent of the country’s agricultural holdings but combine the most fragmented holdings (3.99 parcels on average), the lowest irrigation coverage (0.284), and the highest insufficiency share (0.711). The Terai belt is the mirror image: its 21 districts hold 48.3 percent of all holdings, operate the most consolidated land (2.57 parcels), irrigate 76.7 percent of their area, and report the lowest insufficiency (0.422). The Hill belt sits between the two on fragmentation and insufficiency but, notably, its irrigation coverage (0.300) is much closer to the Mountain than to the Terai level, reflecting the dependence of both upland belts on smaller farmer-managed systems rather than on large gravity-fed canals.

Table 4.4

Ecological-Belt Profile of Fragmentation, Irrigation, and Insufficiency

Belt	Districts	Holdings share	Parcels	Irrig. share	Insuff. share	Severe share
Mountain	16	0.086	3.989	0.284	0.711	0.018
Hill	40	0.431	2.840	0.300	0.662	0.042
Terai	21	0.483	2.566	0.767	0.422	0.060

Note: holdings-weighted belt means using district holdings as weights. “Severe share” is the share of holdings reporting 10–12 months of food insufficiency. Source: author’s computations from the NSCA 2021/22 district tables.

Two features of the table matter for the analysis that follows. First, the irrigated-area share in the Terai is roughly two and a half times the Hill level while the insufficiency share is 24 percentage points lower, which is the geographic alignment that makes irrigation and belt location so difficult to separate statistically; it motivates both the belt fixed-effects robustness checks and the conditional reading of the irrigation coefficient in Section 4.4.3. Second, the severe 10–12-month share runs in the opposite direction, from 0.018 in the Mountain belt to 0.060 in the Terai: the belts with the least production insufficiency contain the largest share of holdings in chronic food deprivation. This

reversal anticipates the finding of Section 4.4.4 that overall and severe insufficiency are distinct dimensions with nearly opposite geographies.

4.3.4 Correlation Analysis

Table 4.5 reports unweighted and holdings-weighted correlations with the insufficiency share, and Figures 4.2 and 4.3 show the two key relationships at district level. Irrigation is by far the strongest correlate: the weighted correlation of the irrigated-area share with insufficiency is -0.856 (-0.751 for the irrigated-holdings share). Fragmentation correlates positively at $+0.396$, average holding size negatively at -0.355 , and the loan share negatively at -0.643 , while the climate-impact share correlates positively at $+0.466$. The fast-market-access share correlates *positively* with insufficiency ($+0.327$ weighted), a counter-intuitive sign examined in Section 4.4.1. The scatter plots make clear that the irrigation relationship is much steeper and tighter than the fragmentation relationship, anticipating the multivariate finding that irrigation remains the dominant correlate once controls are added.

Table 4.5
Correlations With the Agricultural Insufficiency Share

Variable	Unweighted correlation	Weighted correlation
Average parcels per holding	0.439	0.396
Average holding size (ha)	-0.310	-0.355
Irrigation area share	-0.824	-0.856
Irrigated holdings share	-0.649	-0.751
Loan share	-0.625	-0.643
Subsidy share	0.045	-0.061
Power tiller share	-0.203	-0.102
Fast market access share	0.167	0.327
Climate impact share	0.427	0.466

Note: weighted correlations use district holdings as weights.

Table 4.6 completes the picture with the weighted pairwise correlations among the model variables themselves. Three intercorrelations matter for the multivariate analysis. Fragmentation and irrigation are negatively correlated (-0.41): the most fragmented districts are also the least irrigated, so their bivariate associations with insufficiency partly overlap. The loan share moves closely with irrigation ($+0.64$), suggesting that agricultural credit follows the commercialised, irrigated districts rather than compensating the rain-

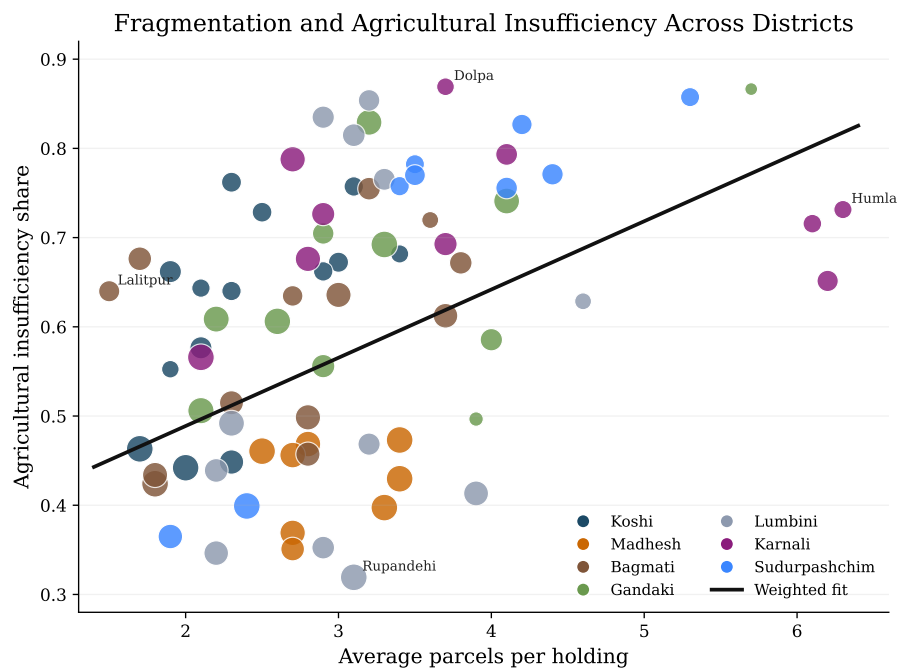
fed ones. And the climate-impact share correlates negatively with irrigation (-0.52) and positively with fragmentation ($+0.41$), so the districts most exposed to adverse climate impacts are also the structurally weakest ones. These overlaps are exactly why the multivariate models of Section 4.4, rather than the bivariate correlations, carry the study's conclusions.

Table 4.6
Weighted Correlation Matrix of the Model Variables

	Frag.	Irrig.	Size	Loan	Subsidy	Mech.	Market	Climate	Insuff.
Frag.	1								
Irrig.	-0.41	1							
Size	-0.12	0.30	1						
Loan	-0.38	0.64	0.38	1					
Subsidy	-0.24	0.05	0.08	-0.12	1				
Mech.	-0.12	0.01	-0.18	0.03	0.13	1			
Market	-0.06	-0.34	-0.45	-0.26	0.09	0.18	1		
Climate	0.41	-0.52	-0.05	-0.37	0.13	-0.12	-0.10	1	
Insuff.	0.40	-0.86	-0.36	-0.64	-0.06	-0.10	0.33	0.47	1

Note: holdings-weighted pairwise correlations across the 77 districts, lower triangle. “Insuff.” is the agricultural insufficiency share (the outcome); the remaining variables are defined in Table 3.1.

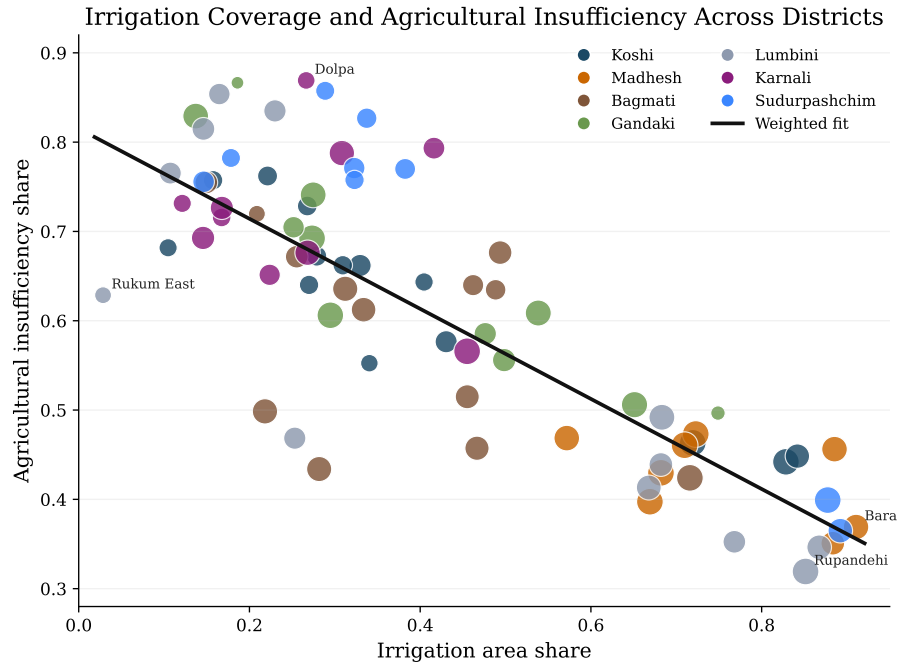
Figure 4.2
Fragmentation and Insufficiency Across Districts



Note: Each point is a district. Colors show provinces and larger circles represent more agricultural holdings.

Figure 4.3

Irrigation and Insufficiency Across Districts



Note: The downward weighted fit line highlights the strong negative district-level association between irrigation and insufficiency.

4.4 The Relationship Between Fragmentation, Irrigation, and Sufficiency

4.4.1 Model Results

Table 4.7 reports the two main specifications: the weighted OLS benchmark and the preferred fractional logit, both with the full control set and province fixed effects. Because fragmentation and irrigation are mean-centred (Section 3.7), the coefficients on the two structural variables measure their associations at the average holding's profile. Three results stand out.

First, irrigation carries the expected negative sign in both models and is estimated precisely: -0.468 ($p < 0.001$) in the weighted OLS and -1.993 ($p < 0.001$) in the fractional logit. Given the degrees-of-freedom constraints discussed in Section 3.6, no single p-value is treated as decisive, and the emphasis throughout is on the stability of signs and magnitudes across specifications. Taking the OLS coefficient at face value, a 10-percentage-point increase in the irrigated-area share is associated with about a 4.7-percentage-point lower insufficiency share, holding the controls and province effects fixed.

Table 4.7*Main Model Results for the Agricultural Insufficiency Share*

	Weighted OLS	Fractional logit
Average parcels per holding	0.0043 (0.0140)	0.0238 (0.0612)
Irrigation area share	-0.4683*** (0.0704)	-1.9932*** (0.3152)
Fragmentation $\times$ irrigation	-0.0287 (0.0525)	-0.1934 (0.2531)
Average holding size (ha)	-0.1411 (0.1419)	-0.5697 (0.6448)
Loan share	0.0419 (0.4209)	0.2113 (1.8216)
Subsidy share	-0.0564 (0.0876)	-0.2712 (0.3654)
Power tiller share	-0.1368 (0.1039)	-0.6728 (0.4592)
Fast market access share	0.4185* (0.2492)	1.8607 (1.1683)
Climate impact share	0.0296 (0.1024)	0.1131 (0.4683)
Constant	0.8398*** (0.1573)	1.4101* (0.7314)
Province fixed effects	Yes	Yes
Observations	77	77
Fit	$R^2 = 0.832$	McFadden $R^2 = 0.058$

Note: robust (HC1) standard errors in parentheses. Both models use normalized district analytical weights based on total holdings. The fractional logit is estimated by quasi-maximum likelihood (Papke & Wooldridge, 1996). Fragmentation and irrigation are mean-centred at their holdings-weighted means (2.81 parcels; 0.525 irrigated-area share) before the interaction is formed, so the main-effect coefficients are evaluated at the average holding's profile. The McFadden pseudo- R^2 is mechanically small for fractional outcomes; the correlation between the fractional logit's fitted and actual district shares is 0.886 (Figure A1, Annex III). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (large-sample normal approximation).

Second, fragmentation is positive in sign, as hypothesised, but imprecisely estimated in both models ($p = 0.76$ and $p = 0.70$), and the interaction between fragmentation and irrigation is negative, directionally consistent with irrigation softening the fragmentation disadvantage, but far from significant ($p = 0.58$ and $p = 0.44$). Section 4.4.3 traces how the strong bivariate fragmentation correlation dissolves as geography is controlled.

Third, among the controls, only the fast-market-access share approaches significance, and it does so with a positive sign (+0.419, $p = 0.093$ in the OLS): districts with better market access report *more* insufficiency, the opposite of the household-level finding of Joshi and Joshi (2017) reviewed in Chapter II. The contradiction is more apparent than real. At district level, fast market access proxies urban and peri-urban composition: places where holdings are small, often part-time, and households meet consumption

through purchases rather than own production. The self-reported production-sufficiency measure is exactly the kind of outcome for which such compositional effects matter, and the coefficient is not robust (it is insignificant in the fractional logit, $p = 0.11$). A direct test supports the compositional reading: adding a market-access $\times$ holding-size interaction to the fractional logit leaves the market-access term positive ($+3.48$, $p = 0.02$) while the interaction enters negatively (-3.97 , $p = 0.19$), so the positive market-access association is concentrated in small-holding districts and fades as average holding size rises, exactly as the urban-composition story predicts, though with 77 districts the interaction is too imprecise to be conclusive. This is a concrete illustration of why district-level associations must not be read as household-level effects.

The province effects themselves are informative: relative to Koshi, only Bagmati's fixed effect is significant (about -0.13 in the OLS, $p = 0.002$), indicating lower insufficiency than its structure would predict, plausibly reflecting the Kathmandu valley's market production and income opportunities.

The fit statistics behave as expected for the two estimator classes. The weighted OLS with province effects explains about 83 percent of the weighted variation in district insufficiency ($R^2 = 0.832$), while the fractional logit's McFadden pseudo- R^2 of 0.058 is mechanically small, as it always is for fractional outcomes whose quasi-likelihood is bounded away from zero. Figure A1 in Annex III shows that the fractional logit's fitted values track the actual district shares closely over their full range, which is the relevant measure of fit for a bounded outcome.

4.4.2 Average Partial Effects and Scenario Predictions

Because fractional logit coefficients are not directly interpretable as marginal changes, Table 4.8 translates the preferred model into average partial effects and scenario predictions, with bootstrap inference from 999 district-level pairs replications. The APE of the irrigated-area share is -0.453 : averaged over the observed district profiles, moving the irrigated share up by one percentage point is associated with insufficiency lower by about 0.45 percentage points. The APE of fragmentation is essentially zero ($+0.005$).

To make these magnitudes concrete, consider a hypothetical district of 50,000 holdings at the weighted-average profile. The irrigation APE implies that an irrigated-area

share higher by 10 percentage points is associated with an insufficiency share about 4.5 percentage points lower, which for such a district amounts to roughly 2,250 fewer holdings reporting that own production cannot cover consumption. Moving across the full interquartile range of irrigation coverage, from 0.230 to 0.668, corresponds to the scenario gap of about 20 percentage points, or some 10,000 holdings in the same hypothetical district.

Table 4.8

Average Partial Effects and Scenario Predictions From the Main Fractional Logit

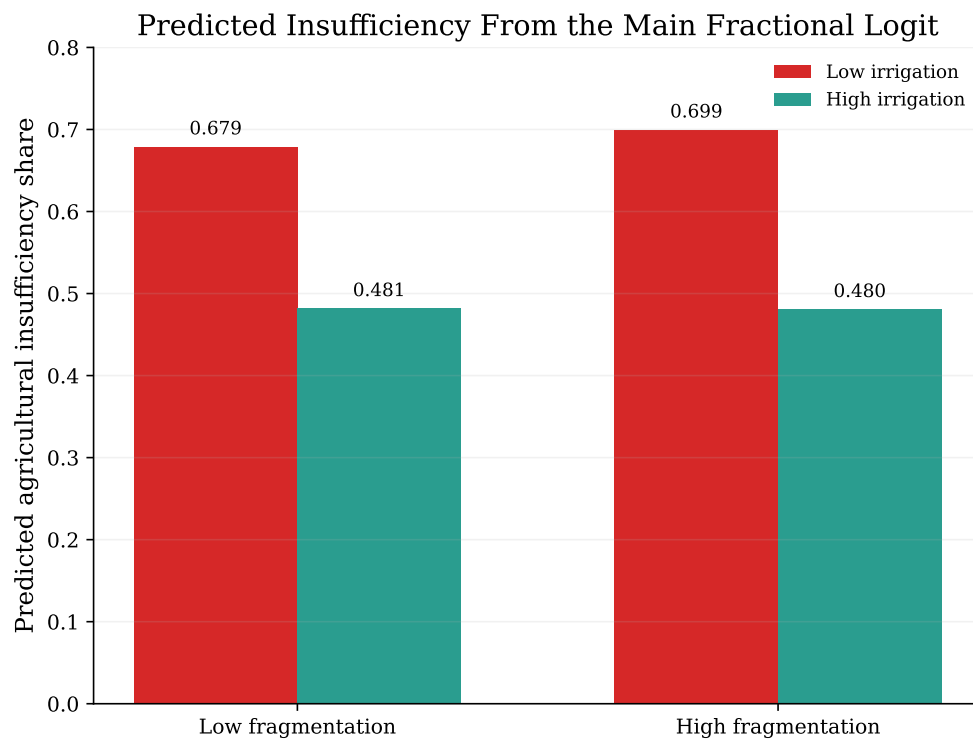
<i>Panel A: average partial effects</i>		
Average partial effect of fragmentation		0.0047
Average partial effect of irrigation area share		-0.4528
<i>Panel B: weighted average predicted insufficiency</i>		
Fragmentation scenario	Irrigation scenario	Prediction
Low fragmentation (2.300)	Low irrigation (0.230)	0.679
Low fragmentation (2.300)	High irrigation (0.668)	0.481
High fragmentation (3.500)	Low irrigation (0.230)	0.699
High fragmentation (3.500)	High irrigation (0.668)	0.480
<i>Panel C: bootstrap inference (999 pairs replications)</i>		
Irrigation coefficient		-1.993 [-2.640, -1.299]
Scenario gap at low fragmentation		-0.197 [-0.266, -0.131]
Scenario gap at high fragmentation		-0.219 [-0.296, -0.128]

Note: scenario gaps are the difference in predicted insufficiency between the high-irrigation (0.668) and low-irrigation (0.230) profiles, computed from the unrounded predictions; they may therefore differ in the last digit from differences of the rounded Panel B values. Brackets contain 95 percent percentile bootstrap confidence intervals from 999 district-level pairs replications.

The scenario panel compares predicted insufficiency at the observed low- and high-irrigation profiles (irrigated shares of 0.230 and 0.668, the lower and upper ends of the interquartile range of the data) for low (2.3) and high (3.5) fragmentation. Moving from the low- to the high-irrigation profile lowers predicted insufficiency from 0.679 to 0.481 at low fragmentation and from 0.699 to 0.480 at high fragmentation, a gap of 19.7 and 21.9 percentage points respectively, with bootstrap 95 percent confidence intervals of $[-0.266, -0.131]$ and $[-0.296, -0.128]$ that exclude zero comfortably. Figure 4.4 visualises the four scenarios. Note also the moderation pattern in the point predictions: the fragmentation gap is about 2 percentage points at low irrigation but only 0.1 at high irrigation, which is descriptively consistent with irrigation neutralising the fragmentation disadvantage, although, as the interaction coefficient shows, the pattern is not statisti-

cally distinguishable from no moderation. The bootstrap interval for the irrigation *coefficient* itself, $[-2.640, -1.299]$, excludes zero, in line with its analytic significance and confirming that the inference does not rest on the large-sample normal approximation. These scenario contrasts are descriptive comparisons of model predictions across observed district profiles; they are not estimates of the effect of an irrigation intervention, and they are not used as such anywhere in this thesis.

Figure 4.4
Scenario Predictions From the Main Fractional Logit



Note: Scenario values come from the main fractional logit using low and high quartile values of fragmentation and irrigation.

4.4.3 Robustness and Specification Checks

Table 4.9 collects the key coefficients from every specification estimated for this thesis. Six lessons emerge.

Table 4.9
Key Coefficients Across All Estimated Specifications

Specification	Fragmentation	Irrigation	Interaction	Fit
Weighted OLS, no FE (baseline)	-0.0060	-0.4242***	-0.0926	0.776
Weighted OLS, province FE	0.0043	-0.4683***	-0.0287	0.832
Fractional logit, province FE (main)	0.0238	-1.9932***	-0.1934	0.058
Fractional logit, irrigated-holdings measure	0.0896	-1.8298***	0.1399	0.053
Weighted OLS, belt FE	-0.0024	-0.3030***	-0.1066*	0.785
Fractional logit, belt FE	-0.0006	-1.2997***	-0.5661*	0.055
Fractional logit, province + belt FE	-0.0069	-1.6326***	-0.2366	0.059
Fractional logit, parsimonious, province FE	0.0319	-2.1437***	-0.2147	0.057
OLS, unweighted, province FE	0.0118	-0.4551***	0.0587	0.799
Fractional logit, unweighted, province FE	0.0484	-1.9529***	0.2103	0.059
Fractional logit, sufficiency outcome	-0.0238	1.9931***	0.1935	0.058
Fractional logit, severe 10–12-month outcome	-0.4499***	0.9672**	-0.5830	0.038
Fractional logit, parsimonious, no interaction	0.0576	-2.0997***	—	0.057
Fractional logit, province FE, valley excluded	0.0385	-2.1102***	-0.2333	0.059
(n=74)				
Fractional logit, household-size control	0.0108	-2.0731***	-0.2238	0.058
Fractional logit, fragmentation as parcels per hectare	0.0503*	-2.0131***	0.0310	0.059

Note: the dependent variable is the agricultural insufficiency share except in the sufficiency model (complement) and the severe model (10–12-month food-insufficiency share). The irrigated-holdings specification replaces the irrigated-area share with the share of holdings reporting irrigation. Fragmentation and irrigation are mean-centred before the interaction is formed, so main-effect coefficients are evaluated at the average holding’s profile. Fit is R^2 for OLS and McFadden pseudo- R^2 for fractional logits, which are not comparable across estimators. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The irrigation association is negative and significant in every specification of the insufficiency outcome. Evaluated at the average holding’s profile, the irrigated-area coefficient is negative with $p < 0.01$ in all fourteen insufficiency-outcome rows of Table 4.9, including the belt-only, parsimonious, and unweighted variants and the two examiner-motivated checks discussed below. Its magnitude does depend on how geography is controlled: relative to the province fixed-effects estimate (−1.993), the coefficient attenuates when ecological-belt dummies *replace* province effects (−1.300) and lies be-

tween the two when province and belt effects enter together (-1.633). The attenuation is consistent with part of the raw irrigation gradient reflecting Nepal's agro-ecological geography, since surface irrigation concentrates in the Terai for reasons of terrain and historical canal investment. The coefficient is therefore read as conditional on the geographic controls, and as a district-level association rather than the causal effect of an irrigation intervention. What survives every specification is both the sign and the policy-relevant magnitude summarised by the scenario gap of Section 4.4.2.

Mean-centring deals with the multicollinearity. Fragmentation, irrigation, and their interaction are mechanically correlated by construction: without centring, the variance inflation factors of the irrigation share and of the interaction reach 17.7 and 13.9, well above the conventional threshold of 10. After the mean-centring described in Section 3.7, every VIF in the main specification is below that threshold (Table 4.10): 2.5 for irrigation, 2.9 for the interaction, and 5.4 for fragmentation. Centring changes no fit, prediction, or interaction estimate; what it changes is where the main effects are evaluated, namely at the average holding's profile rather than at the out-of-sample point of zero fragmentation, which is why the centred irrigation coefficient is estimated far more precisely than its uncentred counterpart. Two further checks confirm that the association is not fragile. Dropping the interaction term altogether in the no-interaction parsimonious specification leaves the irrigation coefficient essentially unchanged (-2.100 , $p < 0.001$, against -2.144 with the interaction), so nothing rests on the interaction term. And re-estimating the main model after excluding the three Kathmandu-valley districts gives -2.110 ($p < 0.001$, $n = 74$), so the result is not the artefact of a few influential urban observations.

It helps to be explicit about what each set of fixed effects absorbs. The province effects hold constant everything the seven provinces share internally, including provincial agricultural policies, extension systems, and budget priorities, while still allowing the Mountain-to-Terai gradient *within* provinces to identify the structural coefficients. The belt effects instead absorb the agro-ecological gradient itself, terrain, climate, and cropping potential, and ask only whether irrigation distinguishes districts *within* the same belt. That the irrigation association survives both, at reduced magnitude under belt controls, is the substantive content of the robustness result.

Table 4.10*Variance Inflation Factors, Main Province Fixed-Effects Specification*

Regressor	VIF
Average parcels per holding	5.39
Irrigation area share	2.46
Fragmentation $\times$ irrigation	2.93
Average holding size (ha)	4.41
Loan share	3.35
Subsidy share	1.52
Power tiller share	1.62
Fast market access share	2.55
Climate impact share	2.27

Note: VIFs are computed for the substantive regressors in the main specification, with province fixed effects partialled out, after fragmentation and irrigation are mean-centred (Section 3.7). All values lie below the conventional threshold of 10, confirming that centring removes the mechanical main-effect/interaction collinearity; without centring, the VIFs of the irrigation share and of the interaction are 17.7 and 13.9.

The bivariate fragmentation disadvantage is absorbed by geography. The weighted correlation of $+0.396$ falls to -0.006 ($p = 0.67$) in the baseline OLS without fixed effects, to $+0.004$ ($p = 0.76$) with province effects, and the APE in the preferred model is $+0.005$. In other words, more fragmented districts do report more insufficiency, but fragmented districts are overwhelmingly mountain and hill districts, and once location is held fixed the independent association of fragmentation is statistically undetectable. The null is informative: it cautions against expecting large sufficiency gains from consolidation alone in unmechanised hill agriculture, echoing the Rwandan evidence of Ali et al. (2019). It should, however, be read with the proxy's limits in mind: average parcels per holding is a count-based proxy that ignores parcel dispersion and inter-plot distance, which are the dimensions the risk-diversification and travel-cost mechanisms of Section 3.3 actually operate on, so part of the attenuation may reflect measurement error in the proxy rather than a genuine absence of association. A partial test is possible with the available data: replacing parcels per holding with parcels per hectare, a proxy that combines parcel count with holding size, leaves the irrigation coefficient intact (-2.013 , $p < 0.001$) while the fragmentation term turns positive at the 10 percent level ($+0.050$, $p = 0.073$; Table 4.9). The richer proxy thus moves the fragmentation association in exactly the direction the measurement-error argument predicts, without reaching conventional significance, so the conjecture that the count-based null understates the true fragmentation relationship is supported but not demonstrated.

The needs-composition concern does not drive the irrigation result. Because the outcome measures production relative to household needs, and needs scale with household size, the district's average household size is the omitted variable of first-order concern for this outcome (Section 1.6). Computed from the census farm-population table (members of holder households divided by holdings; weighted mean 4.71, district range 3.1 to 6.5), and added to the main fractional logit, it changes essentially nothing: the irrigation coefficient moves from -1.993 to -2.073 ($p < 0.001$), and the household-size term itself is small and insignificant ($+0.057$, $p = 0.63$). The composition effect that the self-reported outcome mixes in is therefore not what the irrigation association is picking up.

The interaction question receives a null answer. The fragmentation–irrigation interaction is negative in the main models, with point predictions that narrow the fragmentation gap as irrigation rises (Section 4.4.2), but it is never significant at the 5 percent level (it reaches only 10-percent significance in the two belt-controlled specifications) and it flips sign in the irrigated-holdings and unweighted variants. With 77 observations the design lacks power to detect moderate interaction effects, so the conclusion is “no detectable moderation” rather than “no moderation.”

The results are not driven by the weighting choice. Unweighted estimates, which treat every district equally rather than weighting by holdings, are close to the weighted ones at the average profile (-0.455 against -0.468 in the OLS and -1.953 against -1.993 in the fractional logit, all $p < 0.001$), confirming that the weighted results are not an artefact of the weighting choice (Solon et al., 2015).

The spatial structure of the data receives a direct diagnostic, as set out in Section 3.6. Using a queen-contiguity matrix built from the official district boundaries, Moran's I on the residuals is -0.104 for the weighted OLS and -0.098 for the fractional logit, against an expectation of -0.013 under no autocorrelation, with two-sided permutation p -values of 0.14 and 0.17 (9,999 permutations). There is thus no evidence of positive spatial autocorrelation, which is the pattern that would make the robust standard errors optimistic; if anything the residuals are mildly negatively autocorrelated. The province fixed effects evidently capture the spatial structure of insufficiency, which hardens the inference reported above.

The remaining checks behave as expected: with the complementary sufficiency share as the outcome, all key coefficients reverse sign exactly; and the alternative irrigation measure (irrigated-holdings share) preserves the association (-1.830 , $p < 0.001$).

4.4.4 The Severe Food-Insufficiency Outcome

The severe outcome (the share of holdings reporting food insufficiency for 10 to 12 months) behaves altogether differently, and the difference is instructive rather than embarrassing for the main analysis. In the fractional logit for the severe share, the irrigation coefficient is *positive* and significant ($+0.967$, $p = 0.028$), and fragmentation's coefficient turns negative and significant (-0.450 , $p < 0.001$; Table 4.9). Both signs are the opposite of the main outcome, and both are consistent with severe deprivation concentrating in urbanised, irrigated districts whose holdings are few-parceled and near-landless.

Table 4.11

Overall Versus Severe Insufficiency: Two Different Geographies

Highest <i>overall</i> insufficiency			Highest <i>severe</i> (10–12 months)		
District	Share	Irrig.	District	Share	Irrig.
Dolpa	0.869	0.266	Kaski	0.163	0.538
Manang	0.866	0.186	Kathmandu	0.161	0.494
Bajura	0.857	0.289	Jhapa	0.129	0.719
Arghakhanchi	0.854	0.165	Sunsari	0.102	0.842
Pyuthan	0.835	0.230	Morang	0.095	0.828

Note: “Irrig.” is the district irrigated-area share. The two rankings barely overlap: overall insufficiency peaks in low-irrigation mountain districts, whereas severe insufficiency concentrates in irrigated, more urbanised districts with many small, part-time holdings.

Table 4.11 shows why: overall and severe insufficiency have almost disjoint geographies. Overall insufficiency peaks in low-irrigation mountain districts (Dolpa, with an insufficiency share of 0.869, has a severe share of only 0.001), whereas the severe share peaks in Kaski (0.163) and Kathmandu (0.161) and in irrigated eastern Terai districts: places with many very small, effectively part-time holdings whose members live primarily from non-agricultural income. The two census constructs measure different things: Table 35 captures whether own production covers household needs for the median farming household, while the extreme duration category of Table 36.2 concentrates among near-landless holdings in urbanised districts. Echoing Gartaula et al. (2017), a holding

can be severely “production-insufficient” yet food-secure through purchases. The belt profile of Section 4.3.3 makes the same point at coarser resolution: the severe share rises from 0.018 in the Mountain belt to 0.060 in the Terai, the mirror image of the overall insufficiency gradient, because the irrigated belts contain the urbanising districts where near-landless, purchase-dependent holdings are concentrated. The severe measure is therefore reported as a distinct dimension of deprivation, and the thesis draws no sufficiency conclusions from it; treating it as a confirmation check for the main outcome would be methodologically wrong in both directions.

4.5 Diagnostic Tests

Before the findings are discussed, this section reports the standard post-estimation diagnostics for the main model, covering heteroscedasticity, the normality of the residuals, and multicollinearity. Because the preferred estimator is a fractional logit estimated by quasi-maximum likelihood, which does not assume homoscedastic or normally distributed errors, the classical diagnostics are computed on the residuals of the weighted OLS benchmark of Table 4.7; they nonetheless inform the reading of both models. Table 4.12 collects the results.

Heteroscedasticity is assessed with the Breusch–Pagan test in its studentized (Koenker) form, which does not itself rely on normal residuals. The test statistic is 30.55 with 15 degrees of freedom ($p = 0.010$), so the null of constant error variance is rejected at the 5 percent level. This is unsurprising for district-level data in which the outcome is a bounded share and districts differ greatly in size, and it is precisely why heteroskedasticity-consistent (HC1) standard errors are reported for every specification in this thesis (Section 3.6); the diagnostic confirms that the robust standard errors are the appropriate choice rather than a precaution.

The normality of the residuals is examined with the Jarque–Bera test. The residuals are close to symmetric and near-mesokurtic (skewness -0.21 , kurtosis 3.51), and the test statistic of 1.42 with two degrees of freedom ($p = 0.491$) does not reject normality. Approximate residual normality supports the small-sample behaviour of the weighted OLS benchmark, although the large-sample inference used throughout does not depend on it.

Multicollinearity is monitored with the variance inflation factors reported in full in Table 4.10. After fragmentation and irrigation are mean-centred (Section 3.7), the largest variance inflation factor among the substantive regressors is 5.39, for average parcels per holding, comfortably below the conventional threshold of 10; the irrigation share and the interaction term, whose uncentred factors reach 17.7 and 13.9, fall to 2.46 and 2.93 after centring. The regressors are therefore sufficiently independent for their coefficients to be estimated and interpreted separately.

Taken together with the Moran’s I test for residual spatial autocorrelation reported in Section 4.4.3, which found no evidence of positive spatial dependence, these diagnostics indicate that the inference in this chapter rests on a sound statistical footing: heteroscedasticity is present and is handled by robust standard errors, the residuals are approximately normal, multicollinearity is controlled by centring, and the residuals are not spatially autocorrelated.

Table 4.12

Regression Diagnostic Tests, Main Insufficiency Model

Diagnostic	Test	Statistic (df)	<i>p</i> -value
Heteroscedasticity	Breusch–Pagan	30.55 (15)	0.010
Normality of residuals	Jarque–Bera	1.42 (2)	0.491
Multicollinearity	Maximum VIF	5.39	—

Note: Diagnostics are computed on the residuals of the weighted OLS benchmark (Table 4.7), with the full control set and province fixed effects. The Breusch–Pagan test is reported in its studentized (Koenker) form, which does not assume normal residuals; a *p*-value of 0.010 indicates heteroscedasticity, which is why heteroskedasticity-consistent (HC1) standard errors are reported for all models regardless. The Jarque–Bera test (skewness -0.21 , kurtosis 3.51) does not reject residual normality. The maximum variance inflation factor across the substantive regressors is 5.39 (well below the conventional threshold of 10; the full set is in Table 4.10), so multicollinearity is not a concern after mean-centring.

4.6 Discussion

The first research question asked how land fragmentation and irrigation access vary across the districts of Nepal. Both display wide, geographically structured variation: fragmentation rises and irrigation falls from the Terai through the hills to the mountains, so the two structural disadvantages accumulate in the same places. Karnali combines the most fragmented and least irrigated agriculture in the country, while Madhesh combines the least fragmented and most irrigated. This geographic alignment is not incident-

tal to the analysis; it is precisely why the multivariate models control for province and ecological-belt location before any structural association is interpreted.

The second research question concerned the pattern of agricultural production sufficiency. A majority (55.0 percent, weighted) of holdings report insufficient own production, so production insufficiency is the national norm rather than the exception. The burden is concentrated in mountain and hill districts: province-weighted means range from 0.430 in Madhesh to 0.695 in Karnali, and district shares from 0.319 (Rupandehi) to 0.869 (Dolpa). The severe 10–12-month food-insufficiency measure, by contrast, concentrates in urbanised and irrigated districts with many near-landless holdings, which is why the thesis treats it as a distinct dimension of deprivation rather than a check on the main outcome.

The third research question asked how fragmentation and irrigation relate to production sufficiency and whether irrigation moderates the fragmentation disadvantage. Irrigation emerges as the dominant negative correlate of insufficiency: evaluated at the average holding's profile, its association is negative and significant in every specification of the insufficiency outcome, and the predicted gap between low- and high-irrigation district profiles is roughly 20 percentage points, with bootstrap intervals that exclude zero. The coefficient attenuates when ecological-belt controls replace province effects, and irrigation placement is not random, so the association is read as conditional on geography and as descriptive rather than causal. Fragmentation's apparent bivariate disadvantage, in contrast, is absorbed by geography once location is controlled, and the moderation hypothesis receives a directionally consistent but statistically null answer. On the structure-versus-management debate in the Nepalese literature (Section 2.2.3), the district evidence thus sides qualifiedly with the management view of Karki et al. (2015): what distinguishes high-insufficiency districts is not parcel structure per se but water access and location, which is also consistent with Morioka and Kondo's (2017) finding that productivity growth improved food security despite fragmentation.

The magnitudes are broadly in line with the international evidence reviewed in Chapter II, once differences in outcome and unit of analysis are respected. The OLS estimate implies that a 10-percentage-point difference in the irrigated-area share is associated with about 4.7 percentage points of insufficiency, an order of magnitude comparable to the

roughly 10 percent reduction in food insecurity that Kafle and Balasubramanya (2022) attribute to irrigation access in Niger, and directionally consistent with the large income and food-expenditure gains estimated for Ethiopian irrigators (Ebrahim & Toy, 2024; Jambo et al., 2021). The fragmentation null contrasts with the substantial household-level penalties found in China and Vietnam (Hao et al., 2023; Phan et al., 2022), but the contrast is informative rather than contradictory: the district average parcel count is a coarse proxy that cannot capture plot dispersion, the ecological-inference problem cautions against equating district-level and household-level relationships, and the Rwandan evidence of Ali et al. (2019) shows that the penalty shrinks precisely in unmechanised settings such as the Nepalese hills.

The findings also map directly onto the policy architecture reviewed in Section 2.3. The Agriculture Development Strategy's targets for expanding year-round irrigation are supported by the robust irrigation association (MoAD, 2015). The consolidation instruments of the National Land Policy 2019 and the Land Use Act 2019 find only conditional support: the evidence favours consolidation bundled with irrigation and mechanisation rather than consolidation alone (GoN, 2019; MoLMCPA, 2019), and farmers' own preference for consolidation bundled with infrastructure (Pun et al., 2024) points the same way.

Stated plainly, then, the contributions of this thesis are two. The first is the nationwide district-level mapping of fragmentation, irrigation, and production sufficiency from the NSCA 2021/22, which no previous study provides. The second is the demonstration that irrigation coverage is a robust conditional correlate of production sufficiency, surviving every specification, control, and diagnostic the data support. The fragmentation–irrigation interaction, though theoretically motivated, returns an honest, power-limited null; it is reported as such and is not part of what the thesis claims to establish.

CHAPTER V

SUMMARY AND CONCLUSIONS

5.1 Introduction

This final chapter draws the thesis together. Section 5.2 summarises the objectives, methods, data, and findings; Section 5.3 states the conclusions and their policy implications; and Section 5.4 offers the recommendations of the study.

5.2 Summary

This thesis set out to examine how land fragmentation and irrigation access are associated with agricultural production sufficiency across the districts of Nepal. Three research questions structured the work: how fragmentation and irrigation vary across the 77 districts, how production sufficiency varies, and how the two structural conditions relate to sufficiency, including whether irrigation moderates the fragmentation disadvantage. The first two questions were answered descriptively, through summary statistics, province and belt profiles, district rankings, and correlations; the third through multivariate weighted OLS and fractional logit models with fixed effects and an extensive robustness battery. Motivated by the international evidence that fragmentation lowers efficiency and, in recent studies, worsens household food security, and that irrigation improves productivity and food security, the study constructed a district-level dataset for all 77 districts from the official tables of the NSCA 2021/22, defined the share of holdings reporting insufficient own production as the outcome, and estimated weighted OLS and fractional logit models with province fixed effects, followed by an extensive battery of robustness checks (ecological-belt controls, mean-centred interactions, a parsimonious specification, unweighted estimation, alternative measures and outcomes, household-size and fragmentation-proxy controls, a spatial-autocorrelation diagnostic, and bootstrap inference). Fragmentation and irrigation were centred at their holdings-weighted means before their interaction was formed, so all main effects are read at the average holding's profile, and every table and figure in the thesis is generated by documented

scripts directly from the published census tables, so that each number can be reproduced from the original source.

The descriptive findings show a structurally constrained agriculture. The average holding operates about 0.54 ha in 2.8 parcels, and 55 percent of holdings report that own production is insufficient for household consumption. The three core variables are geographically aligned: fragmentation rises, and irrigation and sufficiency fall, from the Terai through the hills to the mountains, with Karnali and Gandaki carrying the heaviest insufficiency burden and Madhesh the lightest.

The multivariate findings identify irrigation as the dominant correlate of lower insufficiency. The weighted correlation is -0.86 , and the conditional association, evaluated at the average holding's profile, is negative and significant at the 1 percent level in every specification of the insufficiency outcome, including the belt-controlled, parsimonious, and unweighted variants, while the predicted insufficiency gap between low- and high-irrigation profiles is about 20 percentage points, with bootstrap intervals excluding zero. Because the coefficient attenuates under ecological-belt controls and irrigation placement is not random, the claim the thesis defends is conditional and descriptive, not causal. Fragmentation's bivariate disadvantage ($+0.40$ correlation), by contrast, is absorbed by location: its conditional association is statistically indistinguishable from zero in every main specification, and the interaction of fragmentation with irrigation is directionally consistent with moderation but statistically null at $n = 77$. Finally, severe 10–12-month food insufficiency has a different geography from overall insufficiency: it concentrates in urbanised, irrigated districts with many near-landless holdings, and it is a distinct dimension of deprivation, not a check on the main outcome.

5.3 Conclusions

This thesis set out to answer a simple question with the most recent nationwide data available: do the structure of landholdings and access to irrigation distinguish the districts of Nepal where farming feeds households from the districts where it does not? Three conclusions follow from the evidence.

The first conclusion concerns water. In the NSCA 2021/22 district data, irrigation access

is the clearest structural correlate of agricultural production sufficiency. The association survives every specification estimated in this thesis: it is negative and significant at the 1 percent level with province fixed effects, with ecological-belt controls, in the parsimonious and unweighted variants, and under an alternative irrigation measure, and it corresponds to a difference of roughly 20 percentage points in predicted insufficiency between low- and high-irrigation district profiles. Because irrigation placement is not random and the analysis is a district-level cross-section, this finding is an association conditioned on geography rather than a causal effect. Even so, water access, together with the agro-ecological location that irrigation in Nepal inevitably reflects, is what most clearly separates the districts where own production covers household needs from the districts where it does not.

The second conclusion concerns land structure, and it is a null result that carries real information. Once location is held fixed, the average number of parcels per holding adds nothing to the explanation of district insufficiency, and no moderation of the fragmentation disadvantage by irrigation can be detected at $n = 77$. The evidence therefore does not support expecting large sufficiency gains from land consolidation alone in unmechanised hill agriculture. At the same time, the count-based fragmentation proxy ignores parcel dispersion and inter-plot distance, so the null is read as a lower bound on the true fragmentation relationship rather than as proof of its irrelevance.

The third conclusion concerns measurement. Self-reported production sufficiency and duration-based severe food insufficiency capture almost disjoint populations: the former peaks in low-irrigation mountain districts, the latter among near-landless holdings in urbanised and Terai districts. Conflating the two would misdirect targeting in both directions, a point this thesis documents explicitly and that future Nepalese food-security research should heed.

These conclusions must be read against the limitations stated in Section 1.6. The design is cross-sectional and district-level, so the associations reported here are not household-level causal effects; the outcome is self-reported; the fragmentation proxy ignores parcel dispersion and distance; and the census reference year partly coincided with COVID-19 disruptions. None of these limitations reverses a conclusion, but each bounds how far the conclusions travel: the irrigation finding motivates targeting, not an impact estimate,

and the fragmentation null cautions against consolidation-first policy rather than proving consolidation worthless.

Taken together, these conclusions portray agricultural production insufficiency in Nepal as primarily a problem of water and location rather than of parcel structure, and they underline that what a census question measures determines which form of deprivation it reveals.

5.4 Recommendations

The recommendations follow from the study's objectives and are phrased conditionally, as the evidence is associational.

- i. From the first and second objectives, the district-level patterns, the recommendation concerns targeting. Fragmentation, low irrigation coverage, and production insufficiency accumulate in the same hill and mountain districts, with Karnali the starkest case, so agricultural support programmes should be prioritised spatially rather than spread uniformly. The belt profile of Section 4.3.3 identifies the Mountain belt, which holds fewer than 9 percent of the country's holdings but reports the highest insufficiency, as the natural priority stratum. Provincial and local governments, which now hold the extension and local-irrigation mandates under federal restructuring, can read their own districts' position on all three conditions directly from the Annex I dataset and use it for such sub-national prioritisation.
- ii. From the third objective, the multivariate relationship, the central recommendation concerns irrigation. The pattern is consistent with prioritising irrigation expansion and year-round water reliability in low-coverage hill and mountain districts, while recognising that part of the observed gradient reflects agro-ecology that policy cannot move. Because upland districts cannot replicate the Terai's gravity-fed canal systems, the realistic instruments there are the rehabilitation and year-round upgrading of existing farmer-managed systems and small-scale lift or solar irrigation, deliberately targeted so that they do not bypass marginal holdings (Shrestha et al., 2023). Land consolidation, in contrast, should be pursued, if at all, as a complement bundled with irrigation and mechanisation rather than

as a stand-alone remedy: this is consistent with the thesis's null fragmentation result, with farmers' own preferences for consolidation bundled with infrastructure (Pun et al., 2024), and with the sequencing implied by the National Land Policy 2019 and the Land Use Act 2019, whose consolidation and land-protection instruments presume complementary investment rather than substitute for it (GoN, 2019; MoLMCPA, 2019).

- iii. From the analysis of the severe food-insufficiency outcome, finally, the recommendation concerns chronic deprivation. Its geography is nearly the opposite of the production-insufficiency geography: programmes targeting chronic food deprivation should look beyond the agricultural mountain districts to near-landless holdings in peri-urban and Terai districts, whose deprivation is invisible to production-based targeting. Production indicators alone would direct such programmes toward the mountains; the severe-insufficiency evidence of Section 4.4.4 shows they need to be complemented with indicators of landlessness and market dependence if they are to reach the deprived households of districts such as Kaski, Kathmandu, and the eastern Terai.

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ANNEX I: DATA SETS

This annex reports the complete district research dataset constructed for the thesis, as required by the department format for secondary-data-based studies. All variables are computed from the published district tables of the National Sample Census of Agriculture 2021/22 (National Statistics Office open-data portal), using the constructions defined in Table 3.1: each share is the number of holdings with the attribute divided by the district's total holdings, the irrigated-area share divides irrigated area by total holding area, and parcels per holding divides total parcels by total holdings. The ecological belt follows the standard Mountain/Hill/Terai classification of districts. The merged dataset and the scripts that produce every table and figure in this thesis are maintained under version control and are available as a reproducible research archive at <https://github.com/SujilDevkota/economics-thesis> (the version underlying this thesis is tagged `thesis-submission`).

Table A1

District Dataset: Core Variables

District	Belt	Holdings	Parcels	Irrig.	Insuff.	Severe
Achham	Hill	45,214	4.400	0.323	0.771	0.020
Arghakhanchi	Hill	42,418	3.200	0.165	0.854	0.073
Baglung	Hill	50,802	3.200	0.137	0.829	0.060
Baitadi	Hill	47,262	4.100	0.146	0.756	0.030
Bajhang	Mountain	35,330	4.200	0.337	0.827	0.032
Bajura	Mountain	25,279	5.300	0.289	0.857	0.010
Banke	Terai	67,885	2.200	0.682	0.439	0.067
Bara	Terai	84,138	2.700	0.911	0.369	0.023
Bardiya	Terai	80,744	2.200	0.867	0.346	0.050
Bhaktapur	Hill	25,983	2.700	0.488	0.635	0.081
Bhojpur	Hill	34,784	2.500	0.268	0.728	0.026
Chitwan	Terai	89,519	1.800	0.716	0.424	0.074
Dadeldhura	Hill	26,984	3.400	0.323	0.758	0.047
Dailekh	Hill	49,646	2.700	0.308	0.788	0.028
Dang	Terai	101,880	2.300	0.683	0.492	0.062
Darchula	Mountain	24,930	3.500	0.178	0.782	0.028
Dhading	Hill	68,780	3.000	0.312	0.636	0.038
Dhankuta	Hill	31,526	2.300	0.270	0.640	0.025
Dhanusha	Terai	105,255	3.400	0.723	0.473	0.044
Dolakha	Mountain	40,894	3.800	0.255	0.672	0.019
Dolpa	Mountain	8,373	3.700	0.266	0.869	0.001
Doti	Hill	39,572	3.500	0.382	0.770	0.019
Gorkha	Hill	60,028	3.300	0.273	0.692	0.037

Table A1*District Dataset: Core Variables (continued)*

District	Belt	Holdings	Parcels	Irrig.	Insuff.	Severe
Gulmi	Hill	55,911	3.100	0.146	0.815	0.040
Humla	Mountain	9,707	6.300	0.121	0.731	0.002
Ilam	Hill	61,874	1.900	0.330	0.662	0.047
Jajarkot	Hill	33,431	3.700	0.146	0.693	0.008
Jhapa	Terai	148,133	1.700	0.719	0.463	0.129
Jumla	Mountain	21,548	6.200	0.224	0.651	0.006
Kailali	Terai	127,016	2.400	0.877	0.399	0.063
Kalikot	Mountain	24,432	4.100	0.416	0.793	0.031
Kanchanpur	Terai	86,170	1.900	0.892	0.365	0.061
Kapilvastu	Terai	84,675	3.900	0.668	0.413	0.024
Kaski	Hill	53,228	2.200	0.538	0.609	0.163
Kathmandu	Hill	50,857	1.700	0.494	0.676	0.161
Kavrepalanchok	Hill	69,941	2.800	0.218	0.499	0.032
Khotang	Hill	38,131	3.000	0.279	0.672	0.007
Lalitpur	Hill	29,991	1.500	0.462	0.640	0.048
Lamjung	Hill	31,912	2.900	0.498	0.556	0.035
Mahottari	Terai	87,062	2.800	0.572	0.469	0.048
Makawanpur	Hill	67,523	1.800	0.282	0.434	0.013
Manang	Mountain	951	5.700	0.186	0.866	0.013
Morang	Terai	145,318	2.000	0.828	0.442	0.095
Mugu	Mountain	10,929	6.100	0.167	0.716	0.008
Mustang	Mountain	2,366	3.900	0.749	0.497	0.015
Myagdi	Hill	21,792	2.900	0.252	0.705	0.022
Nawalparasi East	Terai	55,412	2.100	0.651	0.506	0.072
Nawalparasi West	Terai	55,098	2.900	0.768	0.352	0.041
Nuwakot	Hill	56,915	2.800	0.467	0.457	0.027
Okhaldhunga	Hill	32,408	3.100	0.157	0.757	0.012
Palpa	Hill	49,742	3.200	0.253	0.468	0.003
Panchthar	Hill	36,718	2.300	0.221	0.762	0.046
Parbat	Hill	25,999	4.000	0.476	0.586	0.027
Parsa	Terai	60,217	2.700	0.883	0.351	0.023
Pyuthan	Hill	50,717	2.900	0.230	0.835	0.057
Ramechhap	Hill	41,863	3.200	0.149	0.755	0.019
Rasuwa	Mountain	9,312	3.600	0.209	0.720	0.021
Rautahat	Terai	86,835	2.700	0.885	0.456	0.032
Rolpa	Hill	46,842	3.300	0.107	0.765	0.011
Rukum East	Hill	11,845	4.600	0.029	0.629	0.004
Rukum West	Hill	32,773	2.900	0.168	0.726	0.009
Rupandehi	Terai	117,333	3.100	0.851	0.319	0.035
Salyan	Hill	49,274	2.800	0.268	0.676	0.019
Sankhuwasabha	Mountain	32,372	2.900	0.309	0.662	0.021
Saptari	Terai	104,660	3.300	0.669	0.397	0.038
Sarlahi	Terai	105,230	2.500	0.710	0.461	0.058
Sindhuli	Hill	54,735	2.300	0.455	0.515	0.041
Sindhupalchok	Mountain	61,865	3.700	0.334	0.612	0.019

Table A1*District Dataset: Core Variables (continued)*

District	Belt	Holdings	Parcels	Irrig.	Insuff.	Severe
Siraha	Terai	104,943	3,400	0.682	0.430	0.045
Solukhumbu	Mountain	22,797	3,400	0.105	0.682	0.006
Sunsari	Terai	97,770	2,300	0.842	0.448	0.102
Surkhet	Hill	62,526	2,100	0.455	0.566	0.035
Syangja	Hill	51,592	4,100	0.275	0.741	0.055
Tanahu	Hill	58,595	2,600	0.295	0.606	0.081
Taplejung	Mountain	24,032	2,100	0.405	0.643	0.020
Terhathum	Hill	18,698	1,900	0.340	0.552	0.012
Udayapur	Hill	61,546	2,100	0.430	0.576	0.052

Table A2*District Dataset: Control Variables*

District	Loan	Subsidy	Power tiller	Fast market	Climate impact
Achham	0.110	0.038	0.000	0.116	0.664
Arghakhanchi	0.070	0.092	0.157	0.214	0.443
Baglung	0.073	0.048	0.245	0.122	0.473
Baitadi	0.061	0.021	0.012	0.131	0.423
Bajhang	0.085	0.026	0.019	0.079	0.580
Bajura	0.055	0.030	0.006	0.121	0.635
Banke	0.137	0.026	0.084	0.103	0.382
Bara	0.168	0.021	0.081	0.128	0.323
Bardiya	0.160	0.044	0.210	0.095	0.406
Bhaktapur	0.043	0.158	0.343	0.410	0.459
Bhojpur	0.074	0.148	0.014	0.133	0.528
Chitwan	0.148	0.052	0.021	0.158	0.361
Dadeldhura	0.097	0.060	0.106	0.199	0.476
Dailekh	0.126	0.028	0.003	0.156	0.443
Dang	0.095	0.030	0.147	0.150	0.311
Darchula	0.036	0.047	0.008	0.122	0.498
Dhading	0.159	0.089	0.109	0.224	0.397
Dhankuta	0.085	0.091	0.093	0.124	0.592
Dhanusha	0.164	0.034	0.051	0.166	0.390
Dolakha	0.076	0.082	0.237	0.216	0.458
Dolpa	0.015	0.005	0.002	0.148	0.283
Doti	0.055	0.023	0.003	0.211	0.207
Gorkha	0.071	0.037	0.097	0.354	0.334

Table A2
District Dataset: Control Variables (continued)

District	Loan	Subsidy	Power tiller	Fast market	Climate impact
Gulmi	0.090	0.143	0.247	0.270	0.532
Humla	0.047	0.025	0.001	0.131	0.567
Ilam	0.141	0.075	0.171	0.086	0.541
Jajarkot	0.100	0.036	0.002	0.104	0.450
Jhapa	0.115	0.421	0.036	0.149	0.432
Jumla	0.062	0.032	0.012	0.108	0.809
Kailali	0.167	0.069	0.498	0.085	0.275
Kalikot	0.132	0.072	0.006	0.072	0.522
Kanchanpur	0.137	0.132	0.244	0.090	0.364
Kapilvastu	0.101	0.052	0.073	0.057	0.470
Kaski	0.067	0.046	0.147	0.213	0.468
Kathmandu	0.107	0.049	0.190	0.360	0.388
Kavrepalanchok	0.138	0.099	0.426	0.230	0.469
Khotang	0.083	0.059	0.023	0.088	0.492
Lalitpur	0.137	0.046	0.068	0.303	0.284
Lamjung	0.072	0.081	0.167	0.222	0.337
Mahottari	0.174	0.021	0.026	0.170	0.367
Makawanpur	0.174	0.030	0.136	0.158	0.401
Manang	0.023	0.560	0.003	0.097	0.840
Morang	0.152	0.046	0.064	0.170	0.262
Mugu	0.064	0.061	0.003	0.159	0.595
Mustang	0.032	0.169	0.017	0.152	0.504
Myagdi	0.056	0.065	0.287	0.171	0.454
Nawalparasi East	0.106	0.083	0.044	0.159	0.313
Nawalparasi West	0.151	0.078	0.095	0.105	0.345
Nuwakot	0.101	0.233	0.282	0.203	0.343
Okhaldhunga	0.161	0.064	0.011	0.098	0.635
Palpa	0.073	0.133	0.383	0.172	0.451
Panchthar	0.081	0.062	0.028	0.181	0.477
Parbat	0.076	0.150	0.380	0.204	0.474
Parsa	0.166	0.032	0.051	0.109	0.337
Pyuthan	0.068	0.122	0.160	0.165	0.377
Ramechhap	0.072	0.063	0.028	0.195	0.322
Rasuwa	0.103	0.112	0.098	0.227	0.219
Rautahat	0.177	0.103	0.042	0.096	0.317

Table A2*District Dataset: Control Variables (continued)*

District	Loan	Subsidy	Power tiller	Fast market	Climate impact
Rolpa	0.075	0.055	0.056	0.095	0.493
Rukum East	0.039	0.014	0.006	0.171	0.398
Rukum West	0.087	0.040	0.033	0.162	0.428
Rupandehi	0.101	0.083	0.037	0.102	0.401
Salyan	0.086	0.064	0.047	0.155	0.365
Sankhuwasabha	0.089	0.059	0.017	0.107	0.607
Saptari	0.126	0.052	0.025	0.144	0.413
Sarlahi	0.185	0.015	0.067	0.126	0.276
Sindhuli	0.102	0.051	0.060	0.143	0.454
Sindhupalchok	0.124	0.076	0.187	0.177	0.375
Siraha	0.144	0.025	0.038	0.128	0.278
Solukhumbu	0.055	0.019	0.008	0.113	0.423
Sunsari	0.144	0.031	0.117	0.108	0.362
Surkhet	0.118	0.046	0.154	0.175	0.348
Syangja	0.051	0.094	0.292	0.187	0.329
Tanahu	0.077	0.099	0.143	0.137	0.347
Taplejung	0.087	0.057	0.009	0.117	0.477
Terhathum	0.091	0.103	0.063	0.113	0.355
Udayapur	0.064	0.026	0.017	0.130	0.349

ANNEX II: MODEL ESTIMATION RESULTS

This annex reports the complete estimation output for the twelve core specifications of this thesis, followed by the fit summary: every coefficient, robust standard error, and approximate p -value, including the province and ecological-belt fixed effects suppressed from the main-text tables. The four supplementary specifications introduced in Section 4.4.3 (the parsimonious model without the interaction term, the Kathmandu-valley-excluded re-estimation, the household-size control, and the parcels-per-hectare fragmentation proxy) are summarised by their key coefficients in Table 4.9. All values are generated directly from the reproducible analysis scripts.

Table A3

Full Estimation Results for All Specifications

Term	Coefficient	Robust SE	p
Weighted OLS, no fixed effects (baseline)			
Constant	0.8475***	0.1068	0.000
Average parcels per holding	-0.0060	0.0139	0.666
Irrigation area share	-0.4242***	0.0684	0.000
Fragmentation $\times$ irrigation (area)	-0.0926	0.0591	0.117
Average holding size (ha)	-0.0630	0.0920	0.493
Loan share	-0.4553	0.4051	0.261
Subsidy share	-0.0959	0.0866	0.268
Power tiller share	-0.1333	0.1160	0.251
Fast market access share	0.1404	0.1862	0.451
Climate impact share	0.0579	0.1185	0.625
Weighted OLS, province fixed effects			
Constant	0.8398***	0.1573	0.000
Average parcels per holding	0.0043	0.0140	0.761
Irrigation area share	-0.4683***	0.0704	0.000
Fragmentation $\times$ irrigation (area)	-0.0287	0.0525	0.585
Average holding size (ha)	-0.1411	0.1419	0.320
Loan share	0.0419	0.4209	0.921
Subsidy share	-0.0564	0.0876	0.519
Power tiller share	-0.1368	0.1039	0.188
Fast market access share	0.4185*	0.2492	0.093

Table A3*Full Estimation Results for All Specifications (continued)*

Term	Coefficient	Robust SE	<i>p</i>
Climate impact share	0.0296	0.1024	0.773
Province FE: Madhesh	-0.0443	0.0401	0.270
Province FE: Bagmati	-0.1284***	0.0421	0.002
Province FE: Gandaki	-0.0383	0.0514	0.456
Province FE: Lumbini	-0.0507	0.0344	0.140
Province FE: Karnali	-0.0434	0.0546	0.427
Province FE: Sudurpashchim	0.0257	0.0536	0.632
Fractional logit, province fixed effects (main)			
Constant	1.4101*	0.7314	0.054
Average parcels per holding	0.0238	0.0612	0.698
Irrigation area share	-1.9932***	0.3152	0.000
Fragmentation $\times$ irrigation (area)	-0.1934	0.2531	0.445
Average holding size (ha)	-0.5697	0.6448	0.377
Loan share	0.2113	1.8216	0.908
Subsidy share	-0.2712	0.3654	0.458
Power tiller share	-0.6728	0.4592	0.143
Fast market access share	1.8607	1.1683	0.111
Climate impact share	0.1131	0.4683	0.809
Province FE: Madhesh	-0.1837	0.1766	0.298
Province FE: Bagmati	-0.5496***	0.1834	0.003
Province FE: Gandaki	-0.1603	0.2337	0.493
Province FE: Lumbini	-0.1965	0.1602	0.220
Province FE: Karnali	-0.1853	0.2523	0.463
Province FE: Sudurpashchim	0.1585	0.2515	0.529
Fractional logit, irrigated-holdings measure			
Constant	0.9990	0.8650	0.248
Average parcels per holding	0.0896	0.0798	0.261
Irrigated holdings share	-1.8298***	0.4627	0.000
Fragmentation $\times$ irrigation (holdings)	0.1399	0.4434	0.752
Average holding size (ha)	-0.3574	0.7612	0.639
Loan share	-1.6895	1.9398	0.384
Subsidy share	-0.4432	0.4578	0.333
Power tiller share	-0.5820	0.4885	0.234

Table A3*Full Estimation Results for All Specifications (continued)*

Term	Coefficient	Robust SE	<i>p</i>
Fast market access share	2.6668*	1.3914	0.055
Climate impact share	1.1473*	0.5984	0.055
Province FE: Madhesh	-0.2273	0.1843	0.217
Province FE: Bagmati	-0.3573*	0.1865	0.055
Province FE: Gandaki	-0.0592	0.2458	0.810
Province FE: Lumbini	-0.2217	0.1893	0.241
Province FE: Karnali	0.0547	0.2656	0.837
Province FE: Sudurpashchim	0.2869	0.2726	0.293
Fractional logit, sufficiency outcome			
Constant	-1.4099*	0.7313	0.054
Average parcels per holding	-0.0238	0.0612	0.697
Irrigation area share	1.9931***	0.3152	0.000
Fragmentation $\times$ irrigation (area)	0.1935	0.2531	0.445
Average holding size (ha)	0.5696	0.6448	0.377
Loan share	-0.2112	1.8215	0.908
Subsidy share	0.2712	0.3654	0.458
Power tiller share	0.6729	0.4592	0.143
Fast market access share	-1.8607	1.1683	0.111
Climate impact share	-0.1132	0.4683	0.809
Province FE: Madhesh	0.1836	0.1766	0.298
Province FE: Bagmati	0.5496***	0.1834	0.003
Province FE: Gandaki	0.1603	0.2337	0.493
Province FE: Lumbini	0.1965	0.1602	0.220
Province FE: Karnali	0.1853	0.2523	0.463
Province FE: Sudurpashchim	-0.1585	0.2515	0.528
Fractional logit, severe 10–12-month outcome			
Constant	-2.2826***	0.8268	0.006
Average parcels per holding	-0.4499***	0.1231	0.000
Irrigation area share	0.9672**	0.4413	0.028
Fragmentation $\times$ irrigation (area)	-0.5830	0.3725	0.118
Average holding size (ha)	-1.1649	0.8148	0.153
Loan share	0.3262	2.5409	0.898
Subsidy share	0.4989	0.5947	0.402

Table A3*Full Estimation Results for All Specifications (continued)*

Term	Coefficient	Robust SE	<i>p</i>
Power tiller share	0.3041	0.5536	0.583
Fast market access share	2.9111**	1.4775	0.049
Climate impact share	0.6532	0.8878	0.462
Province FE: Madhesh	-0.2083	0.2933	0.478
Province FE: Bagmati	-0.6839***	0.2351	0.004
Province FE: Gandaki	0.0568	0.3300	0.863
Province FE: Lumbini	-0.2936	0.2387	0.219
Province FE: Karnali	-0.9082***	0.2979	0.002
Province FE: Sudurpashchim	-0.4721	0.3643	0.195
Weighted OLS, belt fixed effects			
Constant	0.7180***	0.1277	0.000
Average parcels per holding	-0.0024	0.0140	0.862
Irrigation area share	-0.3030***	0.1037	0.004
Fragmentation $\times$ irrigation (area)	-0.1066*	0.0614	0.083
Average holding size (ha)	-0.0174	0.0909	0.848
Loan share	-0.3661	0.4009	0.361
Subsidy share	-0.0749	0.0990	0.449
Power tiller share	-0.1617	0.1201	0.178
Fast market access share	0.1193	0.1911	0.532
Climate impact share	0.0153	0.1229	0.901
Belt FE: Mountain	0.0854*	0.0502	0.089
Belt FE: Hill	0.0795*	0.0469	0.090
Fractional logit, belt fixed effects			
Constant	0.9179	0.5708	0.108
Average parcels per holding	-0.0006	0.0608	0.992
Irrigation area share	-1.2997***	0.4448	0.004
Fragmentation $\times$ irrigation (area)	-0.5661*	0.2950	0.055
Average holding size (ha)	-0.0485	0.3978	0.903
Loan share	-1.6043	1.6846	0.341
Subsidy share	-0.3660	0.4160	0.379
Power tiller share	-0.7273	0.5435	0.181
Fast market access share	0.4936	0.8720	0.571
Climate impact share	0.0643	0.5555	0.908

Table A3*Full Estimation Results for All Specifications (continued)*

Term	Coefficient	Robust SE	<i>p</i>
Belt FE: Mountain	0.3279	0.2197	0.136
Belt FE: Hill	0.3163	0.1987	0.111
Fractional logit, province + belt fixed effects			
Constant	0.9866	0.7800	0.206
Average parcels per holding	-0.0069	0.0620	0.911
Irrigation area share	-1.6326***	0.4152	0.000
Fragmentation $\times$ irrigation (area)	-0.2366	0.2897	0.414
Average holding size (ha)	-0.4490	0.6581	0.495
Loan share	1.0575	1.9333	0.584
Subsidy share	-0.0595	0.4208	0.888
Power tiller share	-0.8238*	0.4805	0.086
Fast market access share	1.9432	1.1922	0.103
Climate impact share	-0.0841	0.4927	0.865
Province FE: Madhesh	-0.0975	0.1797	0.588
Province FE: Bagmati	-0.5681***	0.2063	0.006
Province FE: Gandaki	-0.0833	0.2440	0.733
Province FE: Lumbini	-0.0657	0.1894	0.729
Province FE: Karnali	-0.1788	0.2632	0.497
Province FE: Sudurpashchim	0.2129	0.2612	0.415
Belt FE: Mountain	0.5028**	0.2504	0.045
Belt FE: Hill	0.2935	0.2207	0.184
Fractional logit, parsimonious, province FE			
Constant	2.1936***	0.4125	0.000
Average parcels per holding	0.0319	0.0626	0.610
Irrigation area share	-2.1437***	0.2404	0.000
Fragmentation $\times$ irrigation (area)	-0.2147	0.2520	0.394
Average holding size (ha)	-1.2715**	0.5199	0.015
Province FE: Madhesh	-0.1282	0.1415	0.365
Province FE: Bagmati	-0.6645***	0.2095	0.002
Province FE: Gandaki	-0.3332	0.2028	0.100
Province FE: Lumbini	-0.3270*	0.1714	0.056
Province FE: Karnali	-0.3861*	0.2339	0.099
Province FE: Sudurpashchim	-0.1527	0.1912	0.424

Table A3*Full Estimation Results for All Specifications (continued)*

Term	Coefficient	Robust SE	<i>p</i>
OLS, unweighted, province fixed effects			
Constant	0.8031***	0.1483	0.000
Average parcels per holding	0.0118	0.0155	0.448
Irrigation area share	-0.4551***	0.0621	0.000
Fragmentation $\times$ irrigation (area)	0.0587	0.0491	0.232
Average holding size (ha)	-0.1430	0.1269	0.260
Loan share	-0.0241	0.4057	0.953
Subsidy share	0.1713	0.1308	0.190
Power tiller share	-0.2029*	0.1040	0.051
Fast market access share	0.3738	0.2355	0.113
Climate impact share	0.0444	0.1047	0.672
Province FE: Madhesh	-0.0315	0.0447	0.482
Province FE: Bagmati	-0.0946**	0.0448	0.035
Province FE: Gandaki	-0.0155	0.0480	0.747
Province FE: Lumbini	-0.0399	0.0389	0.305
Province FE: Karnali	-0.0058	0.0524	0.911
Province FE: Sudurpashchim	0.0601	0.0497	0.227
Fractional logit, unweighted, province FE			
Constant	1.2695*	0.7057	0.072
Average parcels per holding	0.0484	0.0683	0.479
Irrigation area share	-1.9529***	0.2865	0.000
Fragmentation $\times$ irrigation (area)	0.2103	0.2311	0.363
Average holding size (ha)	-0.5869	0.5888	0.319
Loan share	-0.0762	1.8698	0.968
Subsidy share	0.7732	0.6579	0.240
Power tiller share	-0.9774**	0.4582	0.033
Fast market access share	1.6531	1.1215	0.141
Climate impact share	0.1966	0.4957	0.692
Province FE: Madhesh	-0.1109	0.2061	0.591
Province FE: Bagmati	-0.4016**	0.1973	0.042
Province FE: Gandaki	-0.0486	0.2219	0.827
Province FE: Lumbini	-0.1414	0.1830	0.440
Province FE: Karnali	-0.0131	0.2504	0.958

Table A3*Full Estimation Results for All Specifications (continued)*

Term	Coefficient	Robust SE	<i>p</i>
Province FE: Sudurpashchim	0.3369	0.2364	0.154

Table A4*Model Fit Summary for All Specifications*

Specification	Fit metric	Value	Converged
Weighted OLS, no fixed effects (baseline)	R^2	0.7764	yes
Weighted OLS, province fixed effects	R^2	0.8322	yes
Fractional logit, province fixed effects (main)	McFadden R^2	0.0582	yes
Fractional logit, irrigated-holdings measure	McFadden R^2	0.0528	yes
Fractional logit, sufficiency outcome	McFadden R^2	0.0582	yes
Fractional logit, severe 10–12-month outcome	McFadden R^2	0.0380	yes
Weighted OLS, belt fixed effects	R^2	0.7850	yes
Fractional logit, belt fixed effects	McFadden R^2	0.0547	yes
Fractional logit, province + belt fixed effects	McFadden R^2	0.0589	yes
Fractional logit, parsimonious, province FE	McFadden R^2	0.0570	yes
OLS, unweighted, province fixed effects	R^2	0.7991	yes
Fractional logit, unweighted, province FE	McFadden R^2	0.0593	yes

ANNEX III: SUPPLEMENTARY GRAPHS

This annex presents three supplementary graphs. Figure A1 plots the fitted against the actual insufficiency shares from the main fractional logit: although the McFadden pseudo- R^2 of the quasi-likelihood model is mechanically small, the fitted values track the actual district shares closely across their full range, which is the relevant measure of fit for a fractional outcome. Figure A2 visualises the near-disjoint geographies of overall and severe insufficiency discussed in Section 4.4.4. Figure A3 shows the concentration of irrigated area in the Terai that underlies the ecological-belt robustness analysis of Section 4.4.3.

Figure A1

Fitted Versus Actual District Insufficiency Shares (Main Fractional Logit)

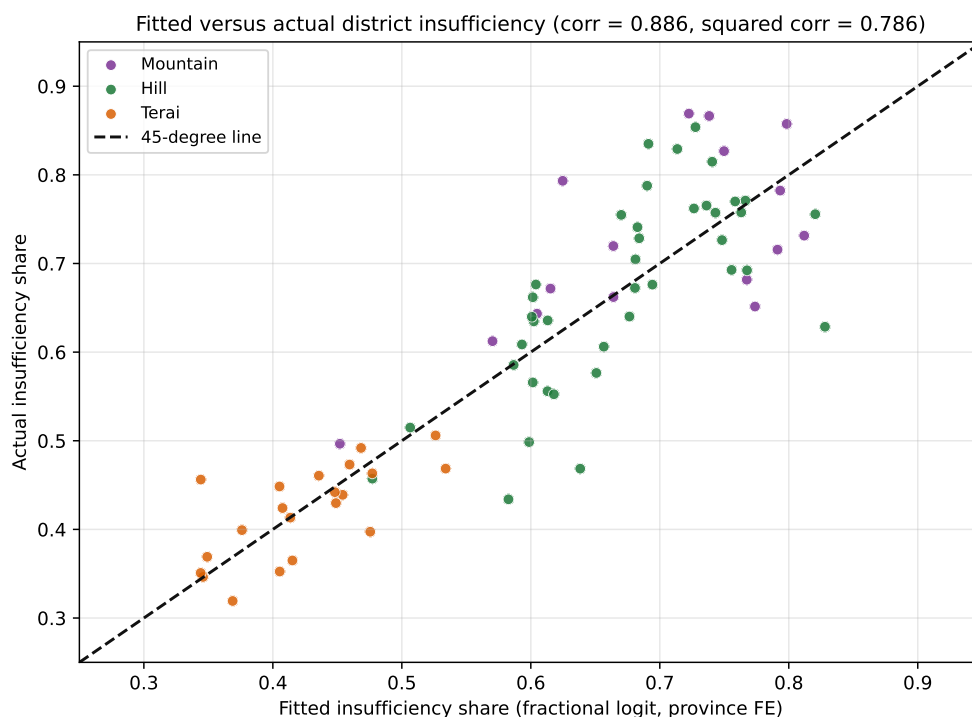


Figure A2

Overall Versus Severe Food Insecurity Across Districts

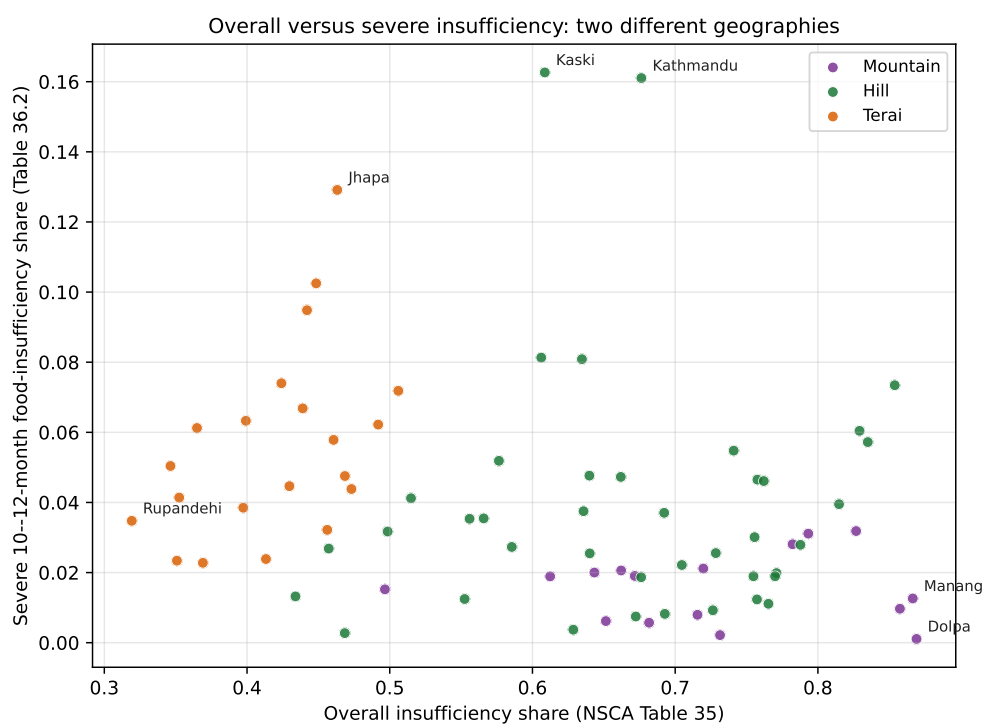


Figure A3

Irrigated-Area Share by Ecological Belt

